Daily Briefing - November 04, 2026

Your Daily Tech & Programming Digest

Wednesday, November 04, 2026

1000	116425	1145	38
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

Automatic computational reproducibility checks with Metacheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-10-30 📖 19 min 📝 3854 words

TWENTY PERCENT STATISTICIAN

> **Summary:** <p>When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

Deidre Artis 2026-09-29 1 min 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

1 min 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D’Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226006949?dgcid=rss_sd_all

Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

A single day of natural daylight exposure and physical activity does not conclusively change circadian phase: A Bayesian crossover experiment

👤 Loock, A.-S., Milanesi, M., Schwendinger, F., Garbazza, C., Schmidt-Trucksaess, A., Lang, U., Blume, C.

📅 2026-11-04 📖 1 min 📝 280 words BIORXIV NEUROSCIENCE

Summary: Among the environmental signals that shape human physiology, the daily light-dark cycle is the primary zeitgeber of the human circadian system. Physical activity may provide an additional non-photic time cue and frequently co-occurs with daylight in everyday life. However, their independent and comb...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.763886v1?rss=1

Recruitment of intrasegmental and propriospinal interneuron networks of forelimb motor circuitry in chronic cervical contusion spinal cord injury

👤 Crowther, K., Harbeck, A., Middlesworth, M., Bosch, M., Lyttle, M. A., MacGibbon, C., Jaffe, D. A., Michel-Flutot, P., Fitchett, E., Li, S., Lepore, A. C.

📅 2026-11-04 📖 1 min 📝 250 words BIORXIV NEUROSCIENCE

Summary: Cervical spinal cord injury (SCI) disrupts descending supraspinal input and spinal networks controlling upper extremity motor function, making recovery of arm and hand function a major priority for affected individuals. The vast majority of regenerative strategies have been studied during the acute ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763629v1?rss=1

Mechanisms of impaired cerebral hemodynamics and memory resilience differ by sex: A role for mineralocorticoid receptors in VCID

DeMarco, G. M., Bernstein, G. A., Chen, D., Bloete, J., Schellinger, E. D., Westbrook, M. E., Baumoel, E. J., Luchetta, J., Chapman, A. C.

2026-11-04 1 min 190 words BIORXIV NEUROSCIENCE

Summary: INTRODUCTION: Chronic hypertension is associated with cerebral hypoperfusion and is a primary risk factor for vascular contributions to cognitive impairment and dementia (VCID). Aldosterone, a key mediator of hypertension, causes cerebrovascular dysfunction via its mineralocorticoid receptor (MR). M...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.31.763340v1?rss=1

Immersion as inference: An active inference account of induced realities and its relation to the two-competitor model

Renz, L., GauthierDickey, K. L., Vibell, J., Francis, A. L., Corona-Strauss, F. I., Strauss, D. J.

2026-11-04 1 min 339 words BIORXIV NEUROSCIENCE

Summary: Immersive technologies are designed for a single target: the human perceptual system. Engineers of virtual and augmented reality have long recognized that a display does not have to replicate the physical world, it only has to provide the inputs that satisfy the senses at the level at which the brai...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.764079v1?rss=1

Differential modulation of the neurocomputational signature of early and late mismatch responses to spoken words by awareness and task relevance

Harenbrock, J., Straube, T., Dellert, T., Schlossmacher, I.

2026-11-04 1 min 249 words BIORXIV NEUROSCIENCE

Summary: Mismatch responses (MMRs), including the mismatch negativity (MMN) and the P3, are a hallmark of the brains capacity to detect unexpected events. They are discussed to be generated by two main processes: adaptation and prediction error (PE). However, MMR research on underlying mechanisms has mostly ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.757859v1?rss=1

Physical sensorimotor contingencies facilitate learning in monkeys

An, S.-J., Gabel, J., Jazayeri, M., Sohn, H.

2026-11-04 1 min 150 words

BIORXIV NEUROSCIENCE

Summary: Animal studies in neuroscience routinely use virtual tasks that deviate significantly from the animals' natural behavioral repertoire. This choice is often necessary for conducting careful and controlled experimentation. However, this approach raises a fundamental question: do virtual tasks expose n...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.01.763380v1?rss=1

One digit too many: brain rhythms and autonomic responses at the limits of working memory capacity

Kotyusov, A. I., Kasanov, D. A., Zaitseva, A. I., Kotchoubey, B., Pavlov, Y. G.

2026-11-04 1 min 250 words

BIORXIV NEUROSCIENCE

Summary: Physiological measures vary with working memory load, but whether they identify when demands exceed a person's working memory capacity remains unclear. We assessed neural and autonomic responses to exceeding digit span by one digit. During an adaptive digit span task, we recorded EEG, pupillometry, ...

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https://www.biorxiv.org/content/10.64898/2026.11.01.763481v1?rss=1

Integrative Transcriptomic Meta-Analysis Reveals SIRT1-SNCA Correlation in Parkinson's Disease

👤 Cimen, H., Sonmez, E., Hotaslier, F.
📅 2026-11-04 📖 1 min 📝 169 words

BIORXIV NEUROSCIENCE

Summary: Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects midbrain dopaminergic neurons without any approved therapy. Candidate molecular targets identified in one model are not always traceable across the postmortem, induced pluripotent stem cell (iPSC), and cell-based syste...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.763624v1?rss=1

The behaviour bends, the code holds: working-memory category information in the multiple-demand network across graded interruption

👤 Leticevscaia, O., Feredoes, E.
📅 2026-11-04 📖 1 min 📝 279 words

BIORXIV NEUROSCIENCE

Summary: Protecting working memory (WM) from interference is essential for goal-directed behaviour, yet its neural mechanisms remain unclear. The multiple-demand (MD) network - a set of frontoparietal regions recruited by diverse cognitive demands - is a candidate system for maintaining WM content under inte...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.763342v1?rss=1

Contributions of Site Heterogeneity in Multi-site Brain-Phenotype Predictions

👤 Chen, G., Jones, M., Gadiyar, I., Satterthwaite, T. D., Chen, A., Vandekar, S.

📅 2026-11-04 📖 1 min

📝 264 words

BIORXIV NEUROSCIENCE

Summary: Machine learning studies of brain-phenotype associations typically summarize accuracy with a single Pearson correlation, yet in harmonized multi-site data this metric is shaped by training-set size, feature and outcome variability, and true effect heterogeneity. We analyzed structural MRI data from ...

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Fine-tuning brain stimulation to fight Parkinson’s disease

📅 2026-11-04 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03168-y

How a once-controversial idea is advancing understanding of Parkinson’s disease

📅 2026-11-04 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03169-x

Reward-related memory benefits cannot be explained by post-encoding stimulus-related thought

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🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-78089-x

The human claustrum tracks uncertainty and prediction error

📅 2026-11-04 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02471-1

Could chemicals contribute to Parkinson’s?

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https://www.nature.com/articles/d41586-026-03170-4

Six highlights from Parkinson’s disease research

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https://www.nature.com/articles/d41586-026-03172-2

Progress accelerates in Parkinson’s treatments

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🔗 **Read full article:**
https://www.nature.com/articles/d41586-026-03165-1

How the brain protects itself from Parkinson’s disease

📅 2026-11-04 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/d41586-026-03171-3

Single-neuron encoding of rapidly learned visual information reshapes human perception

📅 2026-11-04 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41593-026-02496-6

Parkinson’s: beyond the brain

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https://www.nature.com/articles/d41586-026-03173-1

Single-neuron encoding of rapidly learned visual information reshapes human perception

Gabriel Kreiman 2026-11-04 1 min 38 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02496-6 The authors show that rapid learning modulates neuronal activity in the human visual cortex, requiring extra processing time to support recog...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02496-6

The human claustrum tracks uncertainty and prediction error

2026-11-04 1 min 54 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02471-1 Single-neuron recordings provide direct evidence that the human claustrum tracks uncertainty and prediction errors during adaptive behavior. ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02471-1

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1925112

Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

Ansel Y. Rodríguez-González 2026-10-07 1 min 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1919962

Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

👤 Bocheng Wang 📅 2026-10-07 📖 1 min 📝 217 words FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1869687

Multi-receptor whole-brain modeling reveals affinity-driven chemoarchitectural mechanism of virtual atomoxetine action

👤 Gustavo Deco 📅 2026-11-04 📖 1 min 📝 219 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Linking molecular mechanisms with large-scale brain dynamics remains a fundamental challenge in computational psychiatry. Receptor-informed whole-brain models have provided mechanistic insights into pharmacological interventions, yet they have largely relied on single-receptor topographies that cann...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1854866

Music with monaural beats for anxiety reduction and mood enhancement in individuals with moderate trait-anxiety

👤 Gianluca Micchi 📅 2026-11-04 📖 1 min 📝 279 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Monaural beats embedded in music are a popular anxiety-relief intervention, yet implementation parameters and efficacy remain poorly understood. Across two studies of online participants with moderate trait anxiety, we tested whether theta-frequency (4–7 Hz) monaural beats in music produce anxiolyti...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1980343

Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

👤 Peng Han 📅 2026-10-22 📖 1 min 📝 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1934032

MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1910261

Reliability-aware multimodal sensor fusion for human motion monitoring with safety-constrained neurorobotic feedback in sports training

Jiaoyu Wang 2026-11-04 1 min 247 words FRONTIERS NEUROROBOTICS

Summary: IntroductionEmbodied intelligent systems require reliable perception and safety-constrained feedback mechanisms for human motion understanding under heterogeneous sensing conditions. In this study, volleyball training is considered as a representative dynamic human-motion scenario rather than the so...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1973789

Deep reinforcement learning–driven model-based adaptive PID strategy for six-wheeled rocker–bogie mobile platform

Hongxue Feng 2026-11-04 1 min 202 words FRONTIERS NEUROROBOTICS

Summary: IntroductionSix-wheel rocker–bogie rovers encounter nonlinear dynamics, variable traction, and wheel slip that can degrade the performance of fixed-gain proportional–integral–derivative (PID) controllers. This study develops a hybrid deep reinforcement learning–PID framework to improve steering and ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1905372

Retraction: Hypothermia impairs glymphatic drainage in traumatic brain injury as assessed by dynamic contrast-enhanced MRI with intrathecal contrast

2026-11-04 1 min 0 words FRONTIERS NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.2046766

TEMPO-taVNS: a multimodal platform optimizing thermal, electrical, and mechanical auricular stimulation

Matthew P. Ward 2026-11-04 1 min 176 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is typically delivered electrically, while alternative stimulation modalities and their potential for personalization remain underexplored. We present TEMPO-taVNS (Thermal, Electrical, and Mechanical Parameter Optimization for taVNS), a human-...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1983186

Baseline cognition accounts for most predictive information on cognitive recovery during rehabilitation after carbon monoxide poisoning

Peihui Liu 2026-11-04 1 min 271 words FRONTIERS NEUROSCIENCE

Summary: BackgroundCognitive change during rehabilitation after carbon monoxide (CO) poisoning may differ from prognosis at acute presentation. We examined whether information recorded in a rehabilitation pathway added predictive information beyond baseline cognition.MethodsThis single-center retrospective c...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1975750

NeuroTest: a benchtop testbed for evaluating sensing-capable electrophysiology and neurostimulation systems

Morrow JK, H Cho, M Benjaber, Denison TJ, Widge AS and Herron JA

2026-11-04 1 min

247 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. The development of increasingly complex implantable neuromodulation devices requires extensive benchtop validation and testing prior to in vivo implementation. Our goal is to enable this in vitro testing by creating a benchtop system that is accessible, easy-to-use, and functions reliably...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb82d

Towards semantic brain-computer interfacing: the graded inventory of semantic triggers (GIST) paradigm

Joshua James Podmore, Moein Radman, Riccardo Poli, Silke Paulmann and Ian Daly

2026-11-04

1 min

299 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Semantic brain-computer interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual p...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb724

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261104101053&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261104101053&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/40800964/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261104101053&v=2.20.1

ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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The Amsler Grid Use Study (AGUS): Amsler Grid Nonadherence is Driven by Forgetfulness and Health Beliefs But Not Age

Lisa Nivison-Smith 2026-11-03 1 min 33 words LOW VISION

Summary: CONCLUSIONS: Amsler grid nonadherence was primarily unintentional and mostly due to forgetfulness. Poor adherence was associated with greater concerns in Amsler grid use, indicating modifiable factors that can be addressed to improve adherence.

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

Harkaran S Rana 2026-11-03 1 min 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

Summary: Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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Effects of balance training with electrical muscle stimulation on ankle joint muscle on sensory reweighting and motor strategy

Naoya Hasegawa 2026-11-03 1 min 60 words LOW VISION

Summary: CONCLUSIONS: EMS-based balance training was associated with the induction of visual reweighting dynamics and adaptive postural strategies involving multiple joints. Specifically, EMS-induced involuntary ankle assistance elicited hip-centered adjustments in postural control, contributing to upper-tru...

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Reproxalap 0.25% in dry eye disease: A meta-analysis of randomized controlled trials

Hashem Abu Serhan 2026-11-04 1 min 62 words LOW VISION

Summary: CONCLUSION: Reproxalap was associated with consistent improvements across key clinical outcomes in DED. Exploratory allergic conjunctivitis findings should be considered hypothesis-generating rather than clinically conclusive. Interpretation of the overall evidence is limited by exclusive industry s...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

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Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton
2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong
2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

👤 Hu Zhao 📅 2026-11-01 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

👤 Qianxiang Zhou
📅 2026-11-02 📖 1 min 📝 57 words

TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh
📅 2026-11-02 📖 1 min 📝 48 words

TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

Yingying Zeng
2026-11-03 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao 📅 2026-11-03 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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First mitogenome-based phylogenetic analysis and divergence time estimation of deep-sea elasipodid sea cucumbers (Holothuroidea: Elasipodida)

Haibin Zhang 2026-11-03 1 min 29 words TDCS TACS TRNS

Summary: CONCLUSIONS: Collectively, this study establishes a foundational mitogenomic dataset and provide a robust scientific foundation for future studies on the temporal divergence and evolutionary history of elasipodid sea cucumbers.

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin 📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

👤 Lam Ghai Lim 📅 2026-11-03 📖 1 min 📝 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

Heather M Joseph
2026-11-04 1 min 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

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taVNS for uterine contraction pain and early recovery after spontaneous vaginal birth: a three-arm randomized controlled trial protocol

Tai-Pin Guo 2026-11-03 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: BACKGROUND: Uterine contraction pain after spontaneous vaginal birth (SVB) hinders early maternal recovery. Transcutaneous auricular vagus nerve stimulation (taVNS), as a safe, and non-invasive alternative therapy, can alleviate pain and improve sleep and emotion. Although a recent trial has confirm...

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Engineering Neural Interface Materials for Brain-Computer Interfaces: Toward Reliable Neural Information and Functional Recovery

Jie Wu 2026-11-04 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Motor dysfunction is a major disabling outcome of neurological disorders, including stroke, spinal cord injury, and cerebral palsy. By establishing direct communication between neural activity and external effectors, brain-computer interfaces (BCIs) offer a promising strategy for neural functional r...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

🔗 **Read full article:**
http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

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http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

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Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

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http://doi.org/10.1037/neu0001093

The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

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Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

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Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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What Python project taught you the most about real world development?

👤 /u/ Sonu_Jana 📅 2026-11-04 📖 1 min 📝 44 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I'm interested in hearing about projects that genuinely improved your programming skills. Not necessarily huge projects, but ones that taught you something valuable <!-- SC_ON -->   submitted by   <a href="https://www.reddit.com/user/Sonu_Jana"> /u...

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Incident with Pull Requests, Git Operations and Issues

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Tell HN: Gemini trains on chat history, even for subscribers (personal accounts)

👤 jameslk 📅 2026-11-04 📖 1 min 📝 218 words HACKER NEWS

> **Summary:** If you have a personal Google AI subscription, all your saved chat history in Gemini can be used for training and there is no way to opt out unless you turn off chat history or use temporary chat sessions. These chat transcripts may even be read by human reviewers. Although Google claims to disco...

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Geoffrey Hinton: it might turn out we were better off without AI

👤 alkyon 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

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Synaptics Appears to Use Legal Threats to Block Open-Source DisplayLink Driver

👤 snvzz 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

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AI is forcing us to pay off 40 years of technical debt

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North Korean fake job interview led to $6.3B Binance heist

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Jaron Lanier on parallels between AI and social media

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US voters roundly reject data centers in ballot measures

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Charitable giving with earned and windfall money: behavioral and ERP responses to donation-risk information

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Transcutaneous auricular vagus nerve stimulation attenuates post-stroke dysphagia by modulating microglial polarization via the endothelial VWF/PI3K-Akt axis

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TDP-43 dysfunction destabilizes glial homeostatic identity in ALS/FTD

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> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Anna E Dale, Rita Sattler

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Cannabidiol’s prosexual effects in sexually active and sexually inhibited male rats

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Gabriela Rodríguez-Manzo, Ana Canseco-Alba, Jorge Rojas-Hernández

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Machine learning of connectome-wide functional connectivity in spinal muscular atrophy

📖 1 min 📝 36 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Yufen Li, Huasong Cai, Mengyu Zhang, Mengzhen Yan, Shasha Lan, Huirong Nie, Wanqing Shen, Cui Yan, Boyan Xu, Yujian Liang, Wen Tang, Yijuan Li, Pei Xiang, Yingqian Chen

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Spectral-spatial modeling of neural couplings in functional connectivity of motor networks

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Hamidreza Abbaspour, Melissa H. Wade, Christopher P. Hurt, Christopher L. Gonzalez, Harrison C. Walker, Zachary T. Irwin

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Impact of acoustic property uncertainty on transcranial ultrasound simulation

👤 Kosciessa, J. Q., Webb, T. D., Martin, E., Treeby, B., Butts Pauly, K., Verhagen, L., Pichardo, S.

📅 2026-11-03 📖 1 min 📝 241 words BIORXIV NEUROSCIENCE

> **Summary:** Subject-specific simulations of wave propagation are commonly used to plan and report acoustic and thermal exposure during transcranial ultrasound (TUS) interventions. Their accuracy depends on acoustic and thermal tissue property assumptions, including sound speed, density, and attenuation. Reporte...

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Facing the waves of negative mood: Trait mindfulness moderates the links between physiology and emotion

Campos, A. M., Batchalli Maruthy, G., Cumming, G., Rodriguez, S., Himes, L., Rypma, B.

2026-11-03 1 min 197 words BIORXIV NEUROSCIENCE

Summary: Trait mindfulness is a dispositional ability characterized by awareness, non-judgment, and acceptance of internal and external experiences. Stress is associated with numerous psychological and physical health concerns including depression and anxiety. Mindfulness-based interventions (MBIs) have cons...

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Epilepsy: epidemiology, risk factors, classification, pathophysiology, clinical features, diagnosis and management

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Editorial: Exploring the role of criticality in neural systems

Tiago L. Ribeiro 2026-11-04 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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Editorial: BCIs for monitoring and treating mental health conditions

Andrei Dragomir 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2039121

The association between dietary protein intake and mental disorders in older adults: a scoping review

Aliza Haslinda Hamirudin 2026-11-04 1 min 280 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMental disorders are increasingly prevalent among older adults, significantly affecting their quality of life and overall wellbeing. Recent evidence suggests that dietary factors, particularly protein intake, may play a role in mental health outcomes. However, the association between dieta...

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Editorial: Women in speech and language 2025

Anastasia M. Raymer 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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Editorial: Understanding consciousness: neural, cognitive, and philosophical perspectives

Joachim Keppler 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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MA-Pose and DMMS-GCN: skeleton-based martial-arts category recognition with edge-device deployment

Shilin Guo 2026-11-04 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: With the development of visual sensor technology and human pose estimation algorithms, skeleton-based action recognition has received extensive attention in sports analysis and training systems. The recognition of martial-arts actions is difficult because different styles can share similar poses whi...

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A purinergic receptor dominance model for microglial state transitions: balancing perturbation and restraint during neuroinflammation

Minke Tang 2026-11-04 1 min 183 words FRONTIERS NEUROSCIENCE

Summary: Microglia show substantial functional heterogeneity during neuroinflammatory responses, yet the upstream mechanisms through which they interpret extracellular perturbation-associated signals and transition between functional states remain incompletely understood. Purinergic signaling is well positio...

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The butyrate-m6A-BDNF axis: a hypothesis of gut-brain signal transduction in PSD

Dajun Gu 2026-11-04 1 min 151 words FRONTIERS NEUROSCIENCE

Summary: A reduction in brain-derived neurotrophic factor (BDNF) protein levels has been reported in post-stroke depression (PSD). However, whether transcript abundance and translation efficiency change in parallel or become uncoupled in PSD remains to be determined. In addition to classical transcriptional ...

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Delineating neural contributions to electroencephalogram-based speech decoding

Motoshige Sato, Eri Hatakeyama, Masakazu Inoue, Yasuo Kabe, Sensho Nobe, Akito Yoshida, Atsushi Yamamoto, Yuya Kita, Mayumi Shimizu, Kenichi Tomeoka, Michael X Cohen and Shuntaro Sasai

2026-10-30 1 min 249 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Speech brain–computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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Diff-ADN: a diffusion-guided artifact denoising network with deterministic residual refinement for EEG

Mudasir Jamil, Muhammad Zulkifal Aziz, Binwen Huang and Xiaojun Yu

2026-10-30 1 min 241 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Physiological artifacts degrade electroencephalographic (EEG) recordings and can affect downstream brain–computer interface (BCI) analysis. This study develops a compact single-channel framework for ocular, muscular, cardiac, and mixed-artifact removal. Approach. The Diffusion-Guided Arti...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

Jianfeng Wu 2026-11-03 1 min 32 words LOW VISION

Summary: CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

Carrie B Dolan 2026-11-03 1 min 67 words LOW VISION

Summary: CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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The Amsler Grid Use Study (AGUS): Amsler Grid Nonadherence is Driven by Forgetfulness and Health Beliefs But Not Age

Lisa Nivison-Smith 2026-11-03 1 min 33 words LOW VISION

Summary: CONCLUSIONS: Amsler grid nonadherence was primarily unintentional and mostly due to forgetfulness. Poor adherence was associated with greater concerns in Amsler grid use, indicating modifiable factors that can be addressed to improve adherence.

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

Harkaran S Rana 2026-11-03 1 min 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

Summary: Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

Qianxiang Zhou 2026-11-02 1 min 57 words TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh
📅 2026-11-02 📖 1 min 📝 48 words

TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

👤 Yingying Zeng
📅 2026-11-03 📖 1 min 📝 62 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

👤 Jinwu Wang
📅 2026-11-03 📖 1 min 📝 30 words

TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin 📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

👤 Lam Ghai Lim 📅 2026-11-03 📖 1 min 📝 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

Qiang Gao
2026-11-03 1 min 69 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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https://pubmed.ncbi.nlm.nih.gov/42977063/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

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Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/may-2026

Monthly Updates [November]

📅 2026-11-01 📖 5 min 📝 1034 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/nov-2026

Show HN: Uncracked City, detective game where you question LLM suspects that lie

👤 passkey1510 📅 2026-11-03 📖 1 min 📝 171 words HACKER NEWS

> **Summary:** To give you some context on the project. I'm a solo founder and I use AI heavily in my work. The idea came from wanting players to discover cities. Each city has its own history, so every case is set in a real city with real public places. The crime and the people are on other hand invented. The ...

🔗 **Read full article:**
https://uncracked.city

C++26 Disapproved

👤 thatxliner 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://andreasfertig.com/blog/2026/11/cpp26-dis-failed-how-does-cpp-survive/ Comments URL: <a href="https://news.ycombinator.com/item?id=50360164">https://news.ycombinator.com/item?id=...

🔗 **Read full article:**
https://andreasfertig.com/blog/2026/11/cpp26-dis-failed-how-does-cpp-survive/

Giant canyon on Mars is still tectonically active

👤 gmays 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://lamont.columbia.edu/news/giant-canyon-mars-still-tectonically-active-today Comments URL: <a href="https://news.ycombinator.com/item?id=50360573">https://news.ycombinator.co...

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https://lamont.columbia.edu/news/giant-canyon-mars-still-tectonically-active-today

Arrows to arrows, categories to queries (2025)

👤 firstbase 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://reasonablypolymorphic.com/blog/arrows-to-arrows/ Comments URL: https://news.ycombinator.com/item?id=50361107 Points: 26 # Co...

🔗 **Read full article:**
https://reasonablypolymorphic.com/blog/arrows-to-arrows/

Distinct cellular and genetic differences in colorectal cancer in young patients

👤 gmays 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.dana-farber.org/newsroom/news-releases/2026/study-reveals-distinct-cellular-and-genetic-differences-in-colorectal-cancer-in-younger-patients">https://www.dana-farber.org/newsroom/news-releases/2026/study-reveals-distinct-cellular-and-genetic-differences-in-colore...

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https://www.dana-farber.org/newsroom/news-releases/2026/study-reveals-distinct-cellular-and-genetic-differences-in-colorectal-cancer-in-younger-patients

Jelly Sheet

👤 memalign 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://mrdoob.github.io/codevember-2026/01/ Comments URL: https://news.ycombinator.com/item?id=50361268 Points: 102 # Comments: 20

🔗 **Read full article:**
https://mrdoob.github.io/codevember-2026/01/

New software package: ERPLAB Studio

👤 Steve Luck 📅 2024-06-12 📖 2 min 📝 449 words ERP BOOT CAMP

> **Summary:** We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it’s a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Recording and slides now available for ERPLAB Studio webinar

👤 Steve Luck 📅 2024-06-28 📖 1 min 📝 30 words ERP BOOT CAMP

> **Summary:** We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

> **Summary:** In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

> **Summary:** IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11493624

A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

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http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

🔗 **Read full article:**
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Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

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http://ieeexplore.ieee.org/document/11361141

Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

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http://ieeexplore.ieee.org/document/11359738

Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

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http://ieeexplore.ieee.org/document/11364153

Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

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http://ieeexplore.ieee.org/document/11359456

Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

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Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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http://ieeexplore.ieee.org/document/11272135

Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

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IEEE Engineering in Medicine and Biology Society

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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http://ieeexplore.ieee.org/document/11741004

Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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http://ieeexplore.ieee.org/document/11741002

Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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http://ieeexplore.ieee.org/document/11743154

Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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http://ieeexplore.ieee.org/document/11741005

Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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http://ieeexplore.ieee.org/document/11741000

Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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Welcoming the Newly Elected EMBS AdCom Members for 2026–2028

👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming the Newly Elected EMBS AdCom Members for 2026–2028 appeared first on IEEE EMBS.

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Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

> **Summary:** The post Welcoming & Congratulating Our Newly Elected EMBS ExCom Members appeared first on IEEE EMBS.

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EMBS AdCom Nominations Open

👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

> **Summary:** Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

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IEEE Pulse Editor-in-Chief Search – Submit an Application!

👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

> **Summary:** The post IEEE Pulse Editor-in-Chief Search – Submit an Application! appeared first on IEEE EMBS.

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Exploratory bone–brain transcriptomic associations and an ODE-based digital-twin framework for fracture healing

📖 1 min 📝 15 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chao Ma, Wei Lu, Ruipeng Guo

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https://www.sciencedirect.com/science/article/pii/S0006899326004592?dgcid=rss_sd_all

A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

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https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

Dynamic multi-scale topology learning and cross-modal graph fusion for autism spectrum disorder diagnosis

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Huadong Zhang, Junyong Zhao, Shaoqi Wu, Krisztián Szigeti, Kellermayer Miklós, Jiande Sun, Jianping Qiao

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007050?dgcid=rss_sd_all

Diagnostic classification of ASD improves with positively sparsified DTI inter-regional correlation and graph-theoretical artificial intelligence

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ravi Ratnaik, Sriram Kumar P., Lavanya Selvaganesh, Sandeep Singh Sengar, Jac Fredo Agastinose Ronickom

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Fast rate coding in heterogeneous spiking networks

Smeets, K., Wang, S., Schmutz, V., Gerstner, W.

2026-11-03 1 min 165 words

BIORXIV NEUROSCIENCE

Summary: A network of spiking neurons is said to operate in the rate coding regime if its dynamics can be accurately described by a continuous rate network. But how can discrete spike-based communication between neurons give rise to continuous rate-based dynamics? Computational studies showing that heterogen...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763048v1?rss=1

Self-Other Neural Representational Distinction is Associated with Distorted Social Predictions in Social Anxiety

Qi, Y., Chen, Y., Hu, Y., Fan, Q., Wang, L., Yang, G., Hein, G., Wu, H.

2026-11-03 1 min 250 words

BIORXIV NEUROSCIENCE

Summary: Background: Social anxiety (SA) is characterized by distorted negative expectations about the future. Because optimism bias is inherently comparative, the extent to which these predictions reflect a fundamental disruption in self- versus other-referential processing remains poorly understood. Method...

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https://www.biorxiv.org/content/10.64898/2026.10.29.762906v1?rss=1

Diagnose Harmonisation (DHARM): An open-source tool for the harmonisation and analysis of batch effects in MRI-derived measures.

👤 Turnbull, J., Bhalerao, G., Nichols, T. E., McCarthy, P., Gillis, G., Blane, J., Zhaoyi Mo, B., Warrington, S., Sotiropoulos, S. N., Besenczi, R., Alfaro-Almagro, F., Smith, S. M., Griffanti, L.

📅 2026-11-03 📖 1 min 📝 321 words BIORXIV NEUROSCIENCE

Summary: Technical differences between scanners, acquisition protocols, machine hardware and processing pipelines can introduce systematic variation into MRI-derived measures. Numerous harmonisation methods have been developed to reduce these effects. However, validation of these methods is often performed u...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763051v1?rss=1

Strategy-dependent causal relationships between large-scale cortical activity and perceptual decisions

👤 Mohammadi, Z., Pillow, J. W., Pinto, L., Glaser, J. I.

📅 2026-11-03 📖 1 min 📝 178 words

BIORXIV NEUROSCIENCE

Summary: Animals can adopt different internally generated strategies to solve decision-making tasks, but whether cortical regions distinctly contribute to decisions under different strategies remains unknown. To address this problem, we analyzed three datasets comprising either cortical widefield imaging, or...

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Differential effects of chronic hypertension and sex on vascular density along the septotemporal axis of the hippocampus in a rat model of VCID

Luchetta, J., Schellinger, E. D., Bolton, Z., DeMarco, G. M., Chapman, A. C.

2026-11-03 1 min

207 words

BIORXIV NEUROSCIENCE

Summary: Background: Chronic hypertension is a major modifiable risk factor for vascular contributions to cognitive impairment and dementia (VCID), yet vascular changes that precede cognitive dysfunction remain poorly understood. This is particularly relevant for the hippocampus, a deep brain structure criti...

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https://www.biorxiv.org/content/10.64898/2026.10.31.763336v1?rss=1

Metabolic complexity in the human brain supports metastable network dynamics and declines in normative ageing

Deery, H., Liang, E., Moran, C., Egan, G. F., Jamadar, S. D.

2026-11-03 1 min 253 words

BIORXIV NEUROSCIENCE

Summary: The adaptive capacity of the brain is tied to the complexity of its neural activity. While entropy of brain signals has been used to characterise this complexity in haemodynamic and electrophysiological systems, it remains unknown whether entropy contributes to the temporal dynamics of the metabolic...

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https://www.biorxiv.org/content/10.64898/2026.11.02.763776v1?rss=1

Norepinephrine inhibits Afferent activation of NTS Catecholamine Neurons in Mice

Fernandes, L. G., Jara, J. I., Appleyard, S. M.

2026-11-03 1 min 180 words

BIORXIV NEUROSCIENCE

Summary: The nucleus tractus solitarius (NTS) integrates visceral sensory input to regulate feeding behavior, with catecholaminergic (CA) and neuropeptide Y (NPY)-expressing neurons playing key roles in energy homeostasis. Recent evidence suggests functional heterogeneity among CA-NTS neurons, including dist...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763765v1?rss=1

A Preventive Effect of Exercise in a Mouse Model of Essential Tremor

Constantopoulos, S., Bergdahl, A., Courtemanche, R.

2026-11-03 1 min 222 words

BIORXIV NEUROSCIENCE

Summary: The effects of exercise have not been widely studied for potential benefits in essential tremor (ET), while this has been growing for other neurodegenerative disorders. This study examines the effects of two types of exercise [high-intensity interval training (HIIT) and resistance training (RT)] on ...

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Chronic hypertension disrupts the blood-brain barrier in the ventral hippocampus in females but not males in a rat model of vascular cognitive impairment

Luchetta, J., DeMarco, G. M., Thaqi, A., Bernstein, G. A., Chavara, S. E., Schellinger, E. D., Chen, D., Chapman, A. C.

2026-11-03 1 min 318 words BIORXIV NEUROSCIENCE

Summary: Background: Chronic hypertension is a leading cause of cerebrovascular disease, ischemic stroke, and cognitive impairment. In addition, anxiety and hypertension often co-exist, however, little is known about the relationship between the two. Understanding how chronic hypertension impacts limbic brai...

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https://www.biorxiv.org/content/10.64898/2026.10.31.763337v1?rss=1

Lesion-Derived Estimated Functional and Structural Connectomes Predict Post-Stroke Recovery Outcomes

Gillies, C., Jamison, K., Boes, A., Kuceyeski, A.

2026-11-03 1 min 278 words BIORXIV NEUROSCIENCE

Summary: Accurate prognosis after stroke remains challenging, limiting the ability to predict individual recovery. While advanced connectome MRI can provide further prognostically-useful information regarding affected brain networks, its impractical nature has given rise instead to lesion-based estimates of ...

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Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang
2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1938329

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt
2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

Jihui Hao 2026-11-03 1 min 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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The Amsler Grid Use Study (AGUS): Amsler Grid Nonadherence is Driven by Forgetfulness and Health Beliefs But Not Age

Lisa Nivison-Smith 2026-11-03 1 min 33 words LOW VISION

Summary: CONCLUSIONS: Amsler grid nonadherence was primarily unintentional and mostly due to forgetfulness. Poor adherence was associated with greater concerns in Amsler grid use, indicating modifiable factors that can be addressed to improve adherence.

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

👤 Harkaran S Rana 📅 2026-11-03 📖 1 min 📝 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

👤 Liu Yang 📅 2026-08-19 📖 1 min 📝 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

👤 Hideaki Onishi 📅 2026-09-25 📖 1 min 📝 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

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Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

👤 Hu Zhao 📅 2026-11-01 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

👤 Qianxiang Zhou
📅 2026-11-02 📖 1 min 📝 57 words

TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh
📅 2026-11-02 📖 1 min 📝 48 words

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Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

👤 Yingying Zeng
📅 2026-11-03 📖 1 min 📝 62 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

Konstantinos Margetis
2026-11-03 1 min 63 words TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

Dandan Wan 2026-10-31 1 min 17 words FNIRS

Summary: CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

Kanghui Li 2026-10-31 1 min 68 words FNIRS

Summary: The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

> **Summary:** CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

> **Summary:** [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin
📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

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Designing a zero-dependency Python scraping SDK: sticking with pure stdlib vs adding httpx?

👤 /u/Solid-Bidl 📅 2026-11-03 📖 1 min 📝 331 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Hi r/Python, I have been maintaining a lightweight Python SDK (parse-everything on PyPI) that wraps an extraction and anti-bot bypass pipeline (handling TLS JA3/JA4 fingerprint impersonation, Cloudflare Turnstile, DataDome, etc., along...

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Introduction to Programming in ATS

👤 ibobev 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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macOS 27 does not support CoreStorage volumes

👤 colejohnson66 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Preparing for the Flipper One Kickstarter and opening pre-orders

👤 user_7832 📅 2026-11-03 📖 1 min 📝 14 words HACKER NEWS

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Pratt parsing is a black box

👤 ibobev 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Show HN: Kairos – Open-source risk and fraud decisioning platform

👤 ayanb9440 📅 2026-11-03 📖 1 min 📝 283 words HACKER NEWS

> **Summary:** Hey HN, Ayan here from Finic. We’re launching Kairos, an open source alternative to Sift, Sardine, and other real-time fraud decisioning platforms. Kairos addresses two of the biggest problems facing traditional decisioning tools: a lack of labeled data to train on and dealing with fraudsters tha...

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https://github.com/finic-ai/kairos

US tech giants' hidden AI debt swells 90% to $3.2T in 6 months

👤 angry_impala 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Safari introduces Adaptive Dark Mode in 27.3

👤 paddingtone 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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The Little Book of Generative AI Foundations: An Intuitive Mathematical Primer

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The Internals of PostgreSQL

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Plasma p-tau181, Aβ1-42, and GFAP across Alzheimer's disease, mixed-type Alzheimer's disease, mild cognitive impairment, anxiety disorders, and depressive disorders

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Haifeng Lou, Chenyun Huang, Ji Shen, Dake Wang

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RAD51 expression in pan-cancer and its prognostic and immune mechanisms in glioma

📖 1 min 📝 35 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Liubing Hou, Yu Wang, Xiang Zhan, Jiayuan Li, Zizhou Zhang, Weijing Cai, Linlin Su, Wei Han, Xuetao Han, Yue Han, Xiaolan Lv, Xiaoying Xue, Huandi Zhou

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Genetic variants in NEF2L2 rs35652124 and GABRB3 rs2081648 genes: association with autism spectrum disorder in Iraqi children

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zainab Mohammed Burhan, Wisam Hindawi Hoidy

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Propofol suppresses intrinsic excitability and alters excitatory synaptic plasticity in ventral tegmental area dopamine neurons

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Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Min Hee Lee, Dong Kwan Kim

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Improved motor learning in adenosine A1 receptor knockout mice

📖 1 min 📝 19 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maxx Freed, Noah Raffone, Devin Wilke, R. Holly Fitch, Maxim Volgushev

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Feasibility of an 8-week lifestyle intervention combining exercise, diet, cognitive training, and social engagement in individuals with mild cognitive impairment: a pilot randomized controlled trial

📖 1 min 📝 26 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Cai-Lian Wang, Hong Chen, Ruo-Yu Xu, Na Wei, Xiao-Qin Su, Jing Li, Ci Li, Jun-E Liu, Yu-Chen Qiao

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Characterization of the hemodynamic response function in default mode network subregions for task-evoked negative BOLD responses

📖 1 min 📝 29 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Hengda He, Bardiya G. Yazdi, Amirreza Sedaghat, Seyed Hani Hojjati, Sindy Ozoria, Peter Chernek, Jenseric Calimag, Farnia Feiz, Qolamreza R. Razlighi

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A multimodal MRI study of DTI-ALPS and cerebral blood flow changes across the Alzheimer's disease continuum

📖 1 min 📝 38 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yian Gao, Manxi Xu, Peiling Ou, Haoyu Yang, Jia Wang, Wulong Liu, Yu Sun, Wenying Du, Yuli Wang, Tongxi Liu, Hui Xia, Liping Fu, Ni Shu, Zhen Yuan, Guolin Ma

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A phase decoder for reconstructing missing muscle activity

👤 Bahdasariants, S., Yakovenko, S. 📅 2026-11-03 📖 1 min 📝 140 words BIORXIV NEUROSCIENCE

Summary: Upper-limb assistive devices are commonly controlled with electromyographic (EMG) signals, but nerve injury, muscle loss, or practical limits on electrode placement can leave only a few usable signals, which constrains the decoding of intended movements. Here, we introduce a decoder that reconstruct...

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State-dependent emergence of coupling between oscillatory and aperiodic neural activity

Lu, Y., Li, Z., Mao, H., He, X., Liu, D., Chen, J., Tao, L., Xiao, Z.-C., Tian, X.

2026-11-03 1 min

150 words

BIORXIV NEUROSCIENCE

Summary: Neural oscillations and aperiodic activity are two quintessential mesoscopic-level features arising from shared circuits, yet what governs their relationship remains unclear. Here we show that coupling between oscillatory frequency and aperiodic exponent is sporadic and state-dependent. Bias-reduced...

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Pubertal timing is a critical dimension of adolescent brain vulnerability

Szakacs, H., Gombos, F., Kringelbach, M. L., Deco, G., Kovacs, I.

2026-11-03 1 min

200 words

BIORXIV NEUROSCIENCE

Summary: Adolescence is a period of rapid brain development, during which biological maturational timing varies substantially between individuals. This variability may drive specific developmental vulnerabilities to adverse experiences with long-term consequences, warranting early identification of at-risk i...

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Paraquat Inhalation Produces Male-Biased Dopaminergic Neuron Loss and Motor Deficits in Mice: Relevance to Parkinsons Disease

👤 Reina-Gonzalez, P. J., Boos, J. R., Abu-Salah, A., Anchan, A., Strazdins, R., Ay, M., Shah, A., Chalupa, D., Richardson, J. J., Sarkar, S., Cory-Slechta, D.

📅 2026-11-03 📖 1 min 📝 247 words BIORXIV NEUROSCIENCE

Summary: Long-term exposure to paraquat (PQ), a widely used herbicide, has been linked to increased risk of Parkinsons Disease (PD) in agricultural workers and rural communities in the United States (US), findings further supported by experimental animal studies. Although banned in many countries across the ...

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Imipramine binds to amyloid-β and extends the survival of Alzheimer's disease patients

Johnson, N. R., Beham, J., Sillau, S., Read, J., Wang, A. C.-J., Ahmed, M. M., Quang, D., Stephenson, D., Henen, M., Vogeli, B., Chial, H. J., Welty, R., Vugmeyster, L., Potter, H.

2026-11-03 1 min 288 words BIORXIV NEUROSCIENCE

Summary: Background Apolipoprotein E (apoE) plays numerous roles in the pathogenic pathway to Alzheimer's disease (AD). We previously performed a high-throughput screen and identified the antidepressant imipramine as an inhibitor of the interaction between apoE and amyloid-{beta} (A{beta}). We also found tha...

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NeuralFieldManifold: Uncovering the Manifold Geometry of Local Field Potentials

Fallah, K., Chen, H. N., Singha, R., Kong, E., Turi, G. F., Losonczy, A., Zabeh, E.

2026-11-03

1 min 283 words BIORXIV NEUROSCIENCE

Summary: Manifold geometry has become a powerful framework for understanding how the brain organizes high-dimensional neural activity into compact, interpretable structure. To date, however, most theoretical and methodological advances in neural manifold analysis have focused on spiking activity, leaving unr...

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https://www.biorxiv.org/content/10.64898/2026.10.28.762631v1?rss=1

Hmox1/CO regulates retina regeneration via modulation of Wnt/β-catenin signalling and induction of vaspb in zebrafish

Sherpa, R. D., Sen, A., Chaudhuri, A. G., Hui, S. P.

2026-11-03 1 min 258 words

BIORXIV NEUROSCIENCE

Summary: Muller glia (MG) in the zebrafish retina retain a latent stem cell capacity that is absent in mammals, yet the injury responsive signals that drive this reprogramming remain incompletely defined. Here, we identify heme oxygenase-1 (Hmox1), the stress-inducible, rate-limiting enzyme of heme catabolis...

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Brain-wide representations of olfactory behavior in C. elegans

Casademunt, H., Zhang, H., Laver, J., Park, C. F., Kim, S., Fredenberg, W., Hung, W., Calarco, J., Zhen, M., Samuel, A. D. T.

2026-11-03 1 min 196 words

BIORXIV NEUROSCIENCE

Summary: Olfactory navigation emerges from dynamic interactions between sensory input, brain activity state and the animal's own movements. In C. elegans, a compact nervous system enables brain-wide imaging in behaving animals exposed to controlled olfactory environments, with an undulatory locomotion that r...

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Graph neural network approach for inferring neuronal connectivity

👤 Gupta, A., Neveu, C. L., Byrne, J. H.

📅 2026-11-03 📖 1 min 📝 177 words

BIORXIV NEUROSCIENCE

Summary: Understanding the connectivity of neuronal networks is essential for understanding brain function. Current anatomical mapping techniques, such as serial section microscopy and viral tracing, are resource-intensive and cannot capture the efficacy of connectivity. A complementary approach is to infer ...

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Noise-referenced multi-shell diffusion-weighted imaging for regional cerebrospinal fluid thermometry

👤 Behler, C., Breakspear, M., Ramadan, S., Dao, T. T., Close, T. G., Vos, S. B., Thienel, R., Giorgio, J., Behler, A., Fay, M.

📅 2026-11-03 📖 1 min 📝 191 words

BIORXIV NEUROSCIENCE

Summary: We developed a multi-shell diffusion-weighted imaging approach for separate ventricular and subarachnoid cerebrospinal fluid thermometry. The approach uses measured image noise to assess agreement among temperature estimates. In 12 participants, temperature was estimated from four non-zero diffusion...

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Targeting MMP9: an antibody strategy for diabetic peripheral neuropathy

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https://www.nature.com/articles/s41574-026-01322-0

Empathy and compassion are comparable between individuals with cervical spinal cord injury and controls, but their association with interoceptive sensibility differs

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https://www.nature.com/articles/s41598-026-78229-3

Repetitive transcranial magnetic stimulation enhance memory in stress-disordered cynomolgus monkeys by altering the electrical activity of neurons

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The brain lymphatic drainage system: structure, physiological functions, and pathological involvement in Parkinson’s disease

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https://www.nature.com/articles/s41531-026-01595-2

Directional blood flow within a shared capillary network links the nucleus of the solitary tract and area postrema in rats

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https://www.nature.com/articles/s42003-026-11128-5

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

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https://www.nature.com/articles/s41593-026-02535-2

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

👤 Yingjie Zhu
📅 2026-11-03 📖 1 min 📝 24 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 03 November 2026; doi:10.1038/s41593-026-02535-2 Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02535-2

FMRI-guided focal transcranial magnetic stimulation of anterior cingulate cortex reduces opioid seeking and restores connectivity

Zilu MaYing DuanHieu NguyenSabrina LinMd Mohaiminul HaqueDanni WangSamantha HoffmanAidan CarneyTaylor ScottEleni VarlasElliot SteinZheng-Xiong XiYavin ShahamHanbing LuYihong YangaNeuroimaging Research Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224bSchool of Electrical Engineering, International University, VNU-HCM, Ho Chi Minh City 71309, Vietnamchttps://ror.org/00waaqh38Viet Nam National University Ho Chi Minh City, Ho Chi Minh City 71309, Vietnamdhttps://ror.org/043mer456Department of Electrical and Electronic Engineering, University of Nebraska-Lincoln, Lincoln, NE 68588eMolecular Targets and Medication Discovery Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224fBehavioral Neuroscience Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224

📅 2026-10-26 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceA major challenge in developing brain stimulation therapies for opioid addiction is identifying which neural circuits to target and tracking how stimulation reshapes relapse-related circuit functi...

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https://www.pnas.org/doi/abs/10.1073/pnas.2619013123?af=R

Editorial: Data-driven brain imaging and signal analysis: methods from acquisition to clinical insight

Zhiliang Wei

📅 2026-11-03 📖 1 min 📝 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2010682

Peripheral physiological decoding of arousal in a boundary-avoidance task: signal validity and individualized control

Anirudh Natarajan and Paul Sajda

2026-11-03 1 min 284 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Scoring well on a task-demand label does not show that a decoder measures arousal. We ask whether peripheral physiology decodes demand better than EEG, whether the index behaves like graded arousal rather than a clock, and what a controller could do with it. Approach. We re-analysed a pub...

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GROND: automated characterization of periodic and rhythmic EEG patterns

Jin Jing, Chenxi Sun, Tianyu Zhang, Matthew Byrd, Alexandra–Maria Tăuţan, Lara Basovic, Peter N Hadar, Marta P Fernandes, Daniel Goldenholz, Jennifer Kim, Aaron F Struck, Sahar F Zafar and M Brandon Westover

2026-11-03 2 min 469 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Periodic discharges (PDs) and rhythmic delta activity (RDA) are common electroencephalographic patterns in critically ill patients that require detailed characterization—including lateralization, spatial localization, and frequency estimation—according to the American Clinical Neurophysio...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb331

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

👤 Tiffany M Powell-Wiley
📅 2026-11-02 📖 1 min 📝 70 words

LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Effectiveness of spectacle lenses with highly aspherical lenslets (HAL) in slowing axial elongation and spherical equivalent refraction progression in myopic and pre-myopic children: a systematic review and meta-analysis

👤 Jiaqi Yao
📅 2026-11-02 📖 1 min 📝 65 words LOW VISION

Summary: CONCLUSION: Current evidence suggests that HAL spectacle lenses may slow axial elongation and SER progression in myopic children. In pre-myopic children, HAL lenses may offer a smaller benefit, particularly with high adherence, but the evidence is limited and of low quality, precluding a strong reco...

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Changes in self-concept and healthy behaviors following an audio instructional program on puberty among teenage girls with visual impairment: a quasi-experimental study

Ashraf Jehad Abuejheisheh

2026-11-02 1 min 51 words LOW VISION

Summary: CONCLUSION: The results of this investigation revealed that visually impaired adolescent girls had low knowledge, self-concept, and menstrual hygiene practices at puberty before the intervention. All measured outcome variables showed a significant improvement after the implementation of the audio in...

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FCM-Encoder: a novel interpretable framework with vision-language reasoning for coronary artery disease and solitary pulmonary nodule classification

Nikolaos I Papandrianos 2026-11-02 1 min 66 words LOW VISION

Summary: Although modern machine-learning models attain strong predictive performance, their clinical adoption remains limited by a lack of transparency. This concern is particularly relevant in multimodal diagnostic pipelines, where the contribution of each modality to the final prediction is more difficult...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

Jihui Hao 2026-11-03 1 min 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

Ehsan Rezayat 2026-10-31 1 min 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

👤 Hu Zhao 📅 2026-11-01 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

👤 Qianxiang Zhou
📅 2026-11-02 📖 1 min 📝 57 words

TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh

📅 2026-11-02 📖 1 min 📝 48 words

TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

👤 Yingying Zeng
📅 2026-11-03 📖 1 min 📝 62 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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Best hosting platform other than Render (Python) - Django project

👤 /u/Ramesh_9686 📅 2026-11-03 📖 1 min 📝 90 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I designed a personal portfolio ,, https://ramesh13.online/ ,, this is my own website ,, it is a heavy website . My HTTPs request bandwidth is greated than 5 GB as i am hosting in free ...

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https://www.reddit.com/r/Python/comments/1xmr1ap/best_hosting_platform_other_than_render_python/

Bear turns 10, and it still feels like day one

👤 psxuaw 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.bear.app/2026/11/bear-turns-10-and-it-still-feels-like-day-one/ Comments URL: <a href="https://news.ycombinator.com/item?id=50353383">https://news.ycombinator.com/item?id=50...

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Show HN: HackerChurn – which SaaS tools HN wants to cancel

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USDB: Scaling Storage for Next-Generation Recommender Systems at ByteDance

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Gitbook Breached

👤 punchmesan 📅 2026-11-03 📖 2 min 📝 407 words HACKER NEWS

> **Summary:** We opened a ticket with support about Gitbook availability issues and got this in reply. We have yet to see any official customer communication or breach notification. Apologies for the formatting, did the best I could as it's a mobile paste. Update: GitBook security incident update Hey <$na...

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Stringman, an $800 open-source room scale gripper robot

👤 nharada 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Ashby (YC W19) Is Hiring EMs

👤 abhikp 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Postman silently auto-installs AI plugins and attempts to share all API data

👤 sidgarimella 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Qwen 3.8 27B on my laptop can interpret model weights, too

👤 theanonymousone 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://simonwillison.net/2026/Nov/3/qwen-can-interpret-model-weights/ Comments URL: https://news.ycombinator.com/item?id=50355251</...

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How to write comparison content that doesn't age like milk

👤 rbaudibert 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://posthog.com/blog/comparison-content-guide Comments URL: https://news.ycombinator.com/item?id=50355365 Points: 12 # Comments: 3

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I use AI (Oct 2026)

👤 citelao 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ben.stolovitz.com/posts/how-use-ai-oct-2026/ Comments URL: https://news.ycombinator.com/item?id=50355767 Points: 9 # Comments: 6...

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CAG repeat length shapes bidirectional functional connectome alterations in spinocerebellar ataxia type 3: a multimodal MRI study

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Xianfei Yang, Haishuo Xia, Yuwei Hu, Can Xu, Xingang Wang, Qingxin Deng, Qian Xie, Limeng Dai, Qifang Yang, Changying Li, Li Gui, Chen Liu

🔗 **Read full article:**
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Coarse motor-state structure across paired preoperative fMRI and intraoperative ECoG in patients with brain tumors

📖 1 min 📝 36 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Jingxi Hu, Jie Ma, Wanwan Guo, Yang Jiao, Jiaqi Yang, Hongwei Wang, Yiheng Yin, Renke He, Gang Cheng, Hongmin Bai, Hairong Zheng, Dong Liang, Jianning Zhang, Zhanli Hu

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Parietal structural and functional differentiation associated with visual tracking across motor participation profiles

📖 1 min 📝 26 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Keying Zhang, Meng Qi, Ling Jiang, Qing Yan, Yuyan Wang, Shuang Bai, Chunmei Cao, Dong Zhang, Siyuan Ma

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Brief Peppermint Odour Exposure and Immediate Free Recall of a Visual Object Array: A Three-Arm Randomised Experiment

👤 Akar, B., Altıntas, T.
📅 2026-11-02 📖 1 min 📝 270 words
BIORXIV NEUROSCIENCE

Summary: Background: Odours reach limbic and prefrontal targets through a uniquely direct pathway and are often claimed to enhance cognition, yet experimental evidence for specific essential oils is mixed and frequently confounded by expectancy. We tested whether a brief inhalation of peppermint or clove ess...

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Myelin Governs Hippocampal Network Function through the Extracellular Space it Occupies, Not Only the Axons it Insulates

👤 Jia, X., Zeng, C., Liu, L., Gao, T., Bi, J., Zeng, Z., Liang, L., Qi, C., Lin, F., Cheng, F.
📅 2026-11-03
📖 1 min 📝 250 words
BIORXIV NEUROSCIENCE

Summary: Myelin, the multilamellar sheath wrapping neuronal axons, enables saltatory conduction. Beyond electrical insulation, myelin occupies volume, and therefore shapes the extracellular space (ECS) through which interstitial fluid flows and neurotransmitter diffuses. How these two roles interact is unkno...

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Path knockout: an analysis tool for connectome-driven circuit discovery

McGrath, R., Hattori, D.
2026-11-03 1 min 207 words BIORXIV NEUROSCIENCE

Summary: Neural circuits in the brain transform sensory information into perceptual, emotional and motor output. The connectome -- a synaptic-resolution wiring diagram of all neurons in a nervous system -- provides a neuroanatomical blueprint for identifying different circuits that mediate specific computati...

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https://www.biorxiv.org/content/10.64898/2026.10.31.763128v1?rss=1

A medial entorhinal connectome reveals compartmentalized microcircuit architecture

Shi, T., Chen, Y., Jin, H., Jiang, H., Chen, X., Xiong, Z., Wu, F., Zhang, R.
2026-11-03 1 min 217 words BIORXIV NEUROSCIENCE

Summary: The medial entorhinal cortex (MEC) transforms multimodal inputs into spatial representations, but the synaptic architecture underlying its computations remains unclear. Here we generated a synapse-resolution connectomic dataset spanning all six layers of mouse MEC and the adjacent parasubiculum with...

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Wolffia globosa -derived nanovesicles as a natural source of bioactive metabolites with neuroprotective potential against methylglyoxal-induced oxidative stress

Khongkla, E., Boonsnongcheep, P., Srisaisup, M., Charuraksa, T.

2026-11-03 1 min 219 words

BIORXIV NEUROSCIENCE

Summary: Plant-derived nanovesicles (PDNVs) have attracted increasing attention for their potential biomedical applications, particularly in oxidative stress-related diseases and neuroprotection. Wolffia globosa, an emerging functional food and potential future food source, is rich in phytochemicals associat...

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https://www.biorxiv.org/content/10.64898/2026.10.31.762905v1?rss=1

Dissociable norepinephrine and dopamine representations during strategic decision-making and reward learning

Aloor, J., Sit, T. P., Geuter, J., Tutt, J. O., Duan, C. A.

2026-11-03 1 min 256 words

BIORXIV NEUROSCIENCE

Summary: Adaptive decision-making in competitive environments requires learning from immediate reward outcomes while tracking longer-timescale information to adjust behavioural strategies and remain unpredictable against others. Dopamine (DA) and norepinephrine (NE) have long been implicated in reward-guided...

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Oscillatory Phase Coding Emerges in Adaptive RNNS for Flexible Working Memory

Lu, K., Chu, T., TUERHONG, A., Sun, Y., Luo, H., Fang, F., WU, S.

2026-11-03 1 min

210 words

BIORXIV NEUROSCIENCE

Summary: Oscillatory phase coding (OPC) is widely observed in neural systems and has been implicated in cognitive functions such as memory and sequence processing. However, its computational role and neural substrates remain largely unclear. Here, we studied OPC by training recurrent neural networks (RNNs) w...

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Multi-scale place fields provide a predictive basis for hierarchical control

Yu, C., Maimon, S. R., Rich, P. D., Eliav, T., Las, L., Lee, A. K., Ulanovsky, N., Lengyel, M.

2026-11-03 1 min

150 words

BIORXIV NEUROSCIENCE

Summary: Hippocampal place cells typically exhibit unimodal spatial selectivity in small enclosures, but express multiple, irregularly spaced fields in environments of naturalistic scale. However, existing predictive-map theories of the hippocampus do not scale well to large environments and do not account f...

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https://www.biorxiv.org/content/10.64898/2026.10.28.762461v1?rss=1

BundleMorphometry: Spatially Resolved White Matter Bundle Shape Analysis Using Nonlinear Streamline-based Registration

👤 Dasanayake, D., Amit, S. N., Chandio, B. Q.

📅 2026-11-03 📖 1 min 📝 203 words

BIORXIV NEUROSCIENCE

Summary: Diffusion MRI tractometry quantifies microstructural properties along white matter (WM) bundles but does not directly measure their geometry, potentially overlooking localized morphological alterations. Here, we introduce BundleMorphometry, a framework for quantifying local WM bundle shape using the...

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Circuit regulation along an activity continuum shapes flexible and stable representations during learning

👤 Yusufi, D., Ma, Y., Chen, L., Wang, G., Malone, T., Zhang, K., Kim, J. H., Shahid, F., Cekada, K., Barcelone, E., Gu, Y.

📅 2026-11-03 📖 1 min 📝 207 words BIORXIV NEUROSCIENCE

Summary: Learning requires neural representations to flexibly incorporate new information while progressively stabilizing to support enduring memories. How neural circuits balance these demands remains unclear. Here, we identify complementary mechanisms supporting representational flexibility and stability i...

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RIPK1/RIPK3 and MLKL differentially control mitophagy and mitochondrial homeostasis in hypoxic-ischaemic encephalopathy

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Large-scale identification of the regulatory potential of SNPs in human neural stem cells

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Advancing human iPSC-derived motor neuron models using glutamatergic modulators for synaptic function studies

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Integrative analysis of brain-age acceleration: health associations and multi-omics profiles

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https://www.nature.com/articles/s41514-026-00557-5

Subacute carbon monoxide exposure alters AgNOR number and area ratio in the cerebral cortex of rats

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https://www.nature.com/articles/s41598-026-77555-w

Aging-related cellular and molecular changes reveal a role for Bglap3+ neurons in MEC spatial cognition

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41467-026-78673-9

Early-life adversity induces social anxiety and cognitive deficits alongside neurobiological changes in rats

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s42003-026-11279-5

Orexin and pupillary modulation for hunger control

👤 Seher KosarEva BraceyNikola GrujicYihui DuDaria Peleg-RaibsteinDenis Burdakovahttps://ror.org/02q2nz798ETH Zürich, Department of Health Sciences and Technology, Neuroscience Center Zürich, Institute for Neuroscience, Schwerzenbach 8603, Switzerland

📅 2026-10-28 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThis study reveals a genetically defined arousal-hunger circuit that enables mental engagement or external sensory input to naturally interrupt the urge to eat. These findings in mice suggest that...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619759123?af=R

Dlx1&2 directly promote expression of Cxcl14 to control synapse development and interneuron survival

👤 Teng GuoSusan LindtnerJia Sheng HuSiavash Fazel DarbandiYurong WangEmily Ling-Lin PaiAthéna R. YpsilantiCindy T. J. van VelthovenAlex S. NordJohn L. R. RubensteinaNina Ireland Laboratory of Developmental Neurobiology, Department of Psychiatry and Behavioral Sciences, University of California San Francisco, San Francisco, CA 94143bDepartment of Neurobiology, Physiology, and Behavior, Center for Neuroscience, University of California Davis, Davis, CA 95618cDepartment of Psychiatry and Behavioral Sciences, Center for Neuroscience, University of California Davis, Davis, CA 95618dhttps://ror.org/03cpe7c52Allen Institute for Brain Science, Seattle, WA 98109

📅 2026-10-28 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDlxgenes control multiple steps in the development of cortical interneurons (CIN), including synapse development and cell survival. Here we explore molecular mechanisms how theDlxgenes promote neo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2616055123?af=R

Lysine crotonylation acts as an epigenetic switch for glutamate neurotransmission and spatial memory

👤 Utsav MukherjeeBudhaditya BasuPratyush SuryavanshiStacy E. BeyerSaaman GhodsiNathan RobillardYann VanrobaeysEric B. TaylorJoseph GlykysTed AbelSnehajyoti Chatterjeeahttps://ror.org/036jqmy94Department of Neuroscience and Pharmacology, Carver College of Medicine, University of Iowa, Iowa City, IA 52242bhttps://ror.org/036jqmy94Iowa Neuroscience Institute, University of Iowa, Iowa City, IA 52242chttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Neuroscience, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Stead Family Department of Pediatrics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Genetics, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Fraternal Order of Eagles Diabetes Research Center, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Department of Molecular Physiology and Biophysics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242

📅 2026-10-28 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAlthough histone acetylation has long been considered a principal regulator of gene expression events underlying memory storage, the role of non-acetyl histone acyl modifications on long-term memo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2608460123?af=R

Forniceal deep brain stimulation improves memory in Rett syndrome mice by promoting the maturation of newborn hippocampal neurons

Qi WangBin TangZhenyu WuHuda Y. ZoghbiJianrong Tangahttps://ror.org/05cz92x43Jan and Dan Duncan Neurological Research Institute at Texas Children's Hospital, Houston, TX 77030bhttps://ror.org/02pttbw34Department of Pediatrics, Division of Neurology, Baylor College of Medicine, Houston, TX 77030chttps://ror.org/02pttbw34Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030dhttps://ror.org/02pttbw34Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030ehttps://ror.org/02pttbw34Department of Neuroscience, Baylor College of Medicine, Houston, TX 77030fhttps://ror.org/02pttbw34HHMI, Baylor College of Medicine, Houston, TX 77030

📅 2026-10-28 📖 1 min 📝 52 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDeep brain stimulation (DBS) has long been known for its efficacy in mitigating movement disorders. More recently, it has shown promise for addressing disorders of learning and memory: for example...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535887123?af=R

Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

Read full article:
https://www.pnas.org/doi/abs/10.1073/pnas.2526984123?af=R

Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob 2026-10-13 1 min 229 words FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

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https://www.frontiersin.org/articles/10.3389/fninf.2026.1778980

Short excitatory projections promote burst activity and modular functional organization in neural networks

Claus C. Hilgetag 2026-11-03 1 min 165 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Physical constraints in the brain favor short, distance-dependent connections. However, short-range projections are not merely a structural compromise; they also serve essential functional roles. Here, we explore networks with balanced excitation and inhibition, demonstrating that short excitatory p...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1965686

Persistent memory through triple-loop consolidation under stochastic unit turnover

👤 Jianwei Lou 📅 2026-11-03 📖 1 min 📝 366 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Dissipative cognitive architectures maintain computation through continuous energy expenditure, where units that exhaust their energy are stochastically replaced with fresh random state. This creates a fundamental challenge: how can persistent, context-specific memory survive when all learnable stat...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1845213

Editorial: Advances in perceptual decision making and brain oscillations

👤 Francesco Di Gregorio 📅 2026-11-03 📖 1 min 📝 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2023605

Editorial: Machine learning integration in computational neuroscience: enhancing neural data decoding and prediction

George Giakos 2026-11-03 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2030505

Exploratory evaluation of electroencephalogram-derived features for classification of mild cognitive impairment in two independent datasets

Makoto Nakane 2026-11-03 1 min 347 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A proper diagnosis of mild cognitive impairment (MCI) is one of the important challenges in the early detection of dementia. For the classification of MCI patients and healthy controls, some electroencephalogram (EEG)-derived indices were calculated and compared. Some studies have been conducted on ...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1940507

Correction: Post-stroke spasticity: advances in network mechanisms and therapeutics

Zhiqi Lin 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2035182

Embodied multimodal human-computer interaction for English language learning: integrating speech, gestures, emotion, and adaptive feedback

Yao Tang 2026-10-14 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: Current English language learning systems are often limited by single-dimensional learning-state perception, insufficient modeling of affective-cognitive characteristics, and delayed or inflexible feedback regulation. To address these limitations, this study proposes an affective-cognitive gated cro...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1920739

Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams 2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1960197

Editorial: Expanding the therapeutic dose range of cannabinoid-based therapy

Babak Baban 2026-11-03 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2028326

From beat to syntax: evolutionary and developmental paths of a shared neural resource for hierarchical processing in music and language

Tao Gong 2026-11-03 1 min 268 words FRONTIERS NEUROSCIENCE

Summary: Children who track a musical beat well tend to have stronger grammatical skills, and brief exposure to a regular rhythm can transiently improve grammatical processing. These findings raise the possibility of a shared underlying mechanism, and with it two questions: where did the link come from, and ...

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Vagus nerve stimulation in obesity: a systematic review and meta-analysis

Matthew A. Schiefer 2026-11-03 1 min 333 words FRONTIERS NEUROSCIENCE

Summary: BackgroundVagus nerve stimulation (VNS) has been investigated as a potential therapy for obesity because of the vagus nerve’s role in satiety, gastrointestinal signaling, autonomic regulation, and metabolic homeostasis. However, studies vary substantially in stimulation site, interface, parameters, ...

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Analysis and improvement of surface electromyography-based American Sign Language hand gesture recognition

Yasuharu Koike 2026-11-03 1 min 243 words FRONTIERS NEUROSCIENCE

Summary: Surface electromyography (sEMG) allows sign language gestures to be recognized from forearm muscle activity without visual access to the hand. Gestures that produce similar forearm activation patterns, however, remain difficult to distinguish. This study examined how these similarities relate to cla...

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Cybertissue neuroprosthetics: a new implantable neuroprosthetics paradigm to achieve seamless lifetime integration

Liang Guo 2026-10-27 1 min 234 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Despite decades of advances in neural interface technology, the long-term instability of implantable neuroprostheses remains a major barrier to achieving reliable lifetime performance. Chronic foreign-body reactions and fibrotic encapsulation progressively degrade the electrode-tissue int...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb442

Combined voltammetry and optical brain imaging: toward high resolution neurochemical measurements

Vladislav Zavialov, Vassiliy Tsytsarev, Anna B Volnova, Raul R Gainetdinov and Mikhail Inyushin

2026-10-27 1 min 316 words JOURNAL NEURAL ENGINEERING

Summary: The question of the mechanisms by which cellular interactions in the brain are converted into integrative behavioral phenomena is closely related to functional localization, and also to the relationship between spatial and temporal resolution when interpreting experimental data. Classical electrophy...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32e

CORRIGENDUM: In silico development and validation of Bayesian methods for optimizing deep brain stimulation parameters (2023 J. Neural Eng. 20 036015)

Sumedh S Nagrale, Ali Yousefi, Theoden I Netoff and Alik S Widge

2026-10-29 1 min 0 words JOURNAL NEURAL ENGINEERING

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32d

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

Oluwatosin Oluwadare 2026-11-02 1 min 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Exploring Spatial Inequities and Livability: A Mixed-Methods Study Using Artificial Intelligence and Community Insights

Rodrigo Siqueira Reis 2026-11-02 1 min 62 words LOW VISION

Summary: Rapid urbanization has intensified health and social inequities through the uneven distribution of infrastructure, services, and public space. In St. Louis, Missouri (USA), these disparities remain geographically concentrated and rooted in long-standing patterns of racial segregation and disinvestme...

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Association Between Quality of Life and Visual Function in Inherited Retinal Disease: A Cross-sectional Study in a Retinal Genetics Centre

Robert E MacLaren
2026-11-02 1 min 27 words LOW VISION

Summary: CONCLUSIONS: Microperimetry is a patient-relevant measure in inherited retinal diseases. This study demonstrates to stakeholders, such as regulatory bodies, how visual function measures translate to patients' quality-of-life.

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

Tiffany M Powell-Wiley
2026-11-02 1 min 70 words LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Effectiveness of spectacle lenses with highly aspherical lenslets (HAL) in slowing axial elongation and spherical equivalent refraction progression in myopic and pre-myopic children: a systematic review and meta-analysis

👤 Jiaqi Yao
📅 2026-11-02 📖 1 min 📝 65 words LOW VISION

Summary: CONCLUSION: Current evidence suggests that HAL spectacle lenses may slow axial elongation and SER progression in myopic children. In pre-myopic children, HAL lenses may offer a smaller benefit, particularly with high adherence, but the evidence is limited and of low quality, precluding a strong reco...

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Changes in self-concept and healthy behaviors following an audio instructional program on puberty among teenage girls with visual impairment: a quasi-experimental study

Ashraf Jehad Abuejheisheh

2026-11-02 1 min 51 words LOW VISION

Summary: CONCLUSION: The results of this investigation revealed that visually impaired adolescent girls had low knowledge, self-concept, and menstrual hygiene practices at puberty before the intervention. All measured outcome variables showed a significant improvement after the implementation of the audio in...

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FCM-Encoder: a novel interpretable framework with vision-language reasoning for coronary artery disease and solitary pulmonary nodule classification

Nikolaos I Papandrianos 2026-11-02 1 min 66 words LOW VISION

Summary: Although modern machine-learning models attain strong predictive performance, their clinical adoption remains limited by a lack of transparency. This concern is particularly relevant in multimodal diagnostic pipelines, where the contribution of each modality to the final prediction is more difficult...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

Jihui Hao 2026-11-03 1 min 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Effect of electrical stimulation on remyelination in preclinical models of multiple sclerosis: A systematic review

👤 Hamdan Hamdan 📅 2026-10-30 📖 1 min 📝 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang 📅 2026-10-30 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

Qianxiang Zhou 2026-11-02 1 min 57 words TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh 📅 2026-11-02 📖 1 min 📝 48 words TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

👤 Tobias Grossmann 📅 2026-10-29 📖 1 min 📝 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964949/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261103064111&v=2.20.1

Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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https://pubmed.ncbi.nlm.nih.gov/42966412/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1JKSd2KF3MGnV7oFVD2g6PNu7rHRFDsLyCNjKkkf4KHBUA3c8P&fc=None&ff=20261103064111&v=2.20.1

State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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EEG feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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https://pubmed.ncbi.nlm.nih.gov/42962828/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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https://pubmed.ncbi.nlm.nih.gov/42963327/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

🔗 **Read full article:**
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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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https://pubmed.ncbi.nlm.nih.gov/42966015/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

🔗 **Read full article:**
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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42972338/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

What is the closest equivalent to AstroJS in the Python ecosystem?

👤 /u/sentientanus69 📅 2026-11-02 📖 1 min 📝 75 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I am aware a full blown clone wouldn't be possible (hard to imagine how islands would work with Python). But the part where Astro is a free form static site framework is pretty neat and I don't think the established players in this space within Python world are as m...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmcpk7/what_is_the_closest_equivalent_to_astrojs_in_the/

Open-Source 5G Ran Platforms: A Dual Perspective on Performance and Capabilities

👤 teleforce 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2608.25820 Comments URL: https://news.ycombinator.com/item?id=50347464 Points: 39 # Comments: 4

🔗 **Read full article:**
https://arxiv.org/abs/2608.25820

Show HN: I built a little browser game to practice Vim commands

👤 dh1011 📅 2026-11-03 📖 1 min 📝 62 words HACKER NEWS

> **Summary:** I made hjkl to help myself remember Vim commands. It’s a little browser game you can play when you want to unwind and get a bit of Vim practice in. There are still a lot of commands I want to add, but I hope someone finds it useful in the meantime :) <hr /> Comments URL: <a href="https:/...

🔗 **Read full article:**
https://hjkl.duy.run

Nvidia – open-source auto-ontology

👤 campervans 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/NVIDIA/auto-ontology Comments URL: https://news.ycombinator.com/item?id=50348962 Points: 91 # Comments: 30

🔗 **Read full article:**
https://github.com/NVIDIA/auto-ontology

uBlock Origin filter to block autoplay videos in TechCrunch

👤 cornfieldlabs 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://cornfieldlabs.github.io/posts/ublock-stop-techcrunch-autoplay-video/post.html Comments URL: <a href="https://news.ycombinator.com/item?id=50349155">https://news.ycombina...

🔗 **Read full article:**
https://cornfieldlabs.github.io/posts/ublock-stop-techcrunch-autoplay-video/post.html

Revolut moves to self-hosted GLM

👤 pretext 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://twitter.com/guilleflorvs/status/2117379073785696395 Comments URL: https://news.ycombinator.com/item?id=50350004 Points: 87 <...

🔗 **Read full article:**
https://twitter.com/guilleflorvs/status/2117379073785696395

Easily download files from the Open Science Framework with Papercheck

noreply@blogger.com (Daniel Lakens)
2025-07-22 3 min 765 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Visuospatial attention modulates audiovisual temporal perception: Context-dependent associations between prestimulus alpha lateralization and perceptual reports

📖 1 min 📝 30 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Zeliang Jiang, Yuqing Zhao, Weicong Ren, Jie Zhang, Xinyuan Liu, Fei Yu, Xingwei An, Wenxiao Zhong, Di Wu, Pan Zhang, Zhijie Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600635X?dgcid=rss_sd_all

The visual cortex and beyond: investigating long-term cortical plasticity following monocular deprivation

📖 1 min 📝 29 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Franziska Wagner, Lukas Grajewski, Alexander Schmidt, Ulrike Teschner, Markus Ullsperger, Kathrin Finke, Christian Gaser, Robert Dahnke, Carsten M. Klingner, Stefan Brodoehl

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600604X?dgcid=rss_sd_all

Distinct neural patterns of valence and arousal in the developing brain: A representational similarity analysis of MEG

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Ioannis Ntoumanis, Christos Papadelis

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006324?dgcid=rss_sd_all

Single-molecule bouncing at a liquid interface governs presynaptic enrichment

👤 Kudo, K., Longfield, S. F., McCann, A. J., Schoovaerts, N., Sanchez-Velasquez, J., Parra-Rivas, L. A., Roy, S., Verstreken, P., Hinde, E., Millard, S. S., Bademosi, A. T., Meunier, F. A.

📅 2026-11-02 📖 1 min 📝 187 words BIORXIV NEUROSCIENCE

> **Summary:** Many cellular processes rely on accumulating highly diffusive cytoplasmic proteins to a specific location within the cell. This spatial enrichment is generally thought to rely on locally immobilizing proteins or slowing diffusion through binding and molecular crowding. Paradoxically, cytoplasmic pro...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.01.763503v1?rss=1

Prolactin Receptor Long Isoform ΔS1 Expression is Reduced in the Dorsal Root Ganglia of Humans with Chronic Pain

He, L., Natarajan, K., Shiers, S. I., Arendt-Tranholm, A., Sayani, R., Jagan, S., Hossain, A., Rosenfield, J., Rodriguez Concepcion, A. J., Dussor, G., Akopian, A. N., Price, T. J.

2026-11-02 1 min 301 words BIORXIV NEUROSCIENCE

Summary: Prolactin receptor (PRLR) isoforms, specifically the short (PRLR-S) and long (PRLR-L), are differentially expressed in pathological conditions and may be involved in the induction and maintenance of pain in females. PRLR sex-dependent actions are well studied in rodent models of chronic pain, but th...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.762326v1?rss=1

EigenBrain maps cortical molecular architecture from functional, geometric, and structural eigenmodes

Wan, B., Du, P., Ferrari, A., Saberi, A., John, A., Yan, J., Liu, X., Hansen, J. Y., Liu, Z.-Q., Hong, S.-J., Carruzzo, F., Kaliuzhna, M., Geffen, T., Katthagen, T., Misic, B., Thompson, P. M., Van De Ville, D., Margulies, D. S., Warrier, V., Bethlehem, R. A. I., Schlagenhauf, F., Kaiser, S., Bernhardt, B., Valk, S. L., Kirschner, M.

2026-11-02 1 min 232 words BIORXIV NEUROSCIENCE

Summary: Neurotransmitter receptor and transporter distributions shape neural communication across the cortex and underpin mechanistic interpretations of brain function. Magnetic resonance imaging (MRI)-derived eigenmodes, low-dimensional spatial patterns that capture the principal axes of cortical organizat...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.758642v1?rss=1

Bone transport induces CGRP release from sensory nerves to promote angiogenesis and heal diabetic foot wounds in rats

2026-11-03 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41598-026-73891-z

Proapoptotic RHG proteins and mitochondria are required for sensory dendrite pruning in Drosophila

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77081-9

An exploratory remote pilot study of environmental risk factors for Lewy body dementia considering sex differences

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41531-026-01603-5

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

Kay A. Robbins 2026-10-12 1 min 0 words FRONTIERS NEUROINFORMATICS

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss 2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

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https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer’s disease analysis

Arshad Ali 2026-10-29 1 min 287 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer’s disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

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Virtual brain models: a roadmap towards effective brain digital twins

Viktor Jirsa 2026-10-29 1 min 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

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A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-10-29 1 min 314 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMotor imagery (MI) can support self-initiated control, whereas brain–computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspended...

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Resolving the apparent paradox of multisensory expertise: coupling priors, common-cause priors, and how to distinguish among them

Pier Luigi Sacco 2026-10-29 1 min 348 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Perceptual expertise is widely assumed to refine multisensory perception, yet the empirical literature appears inconsistent. Trained musicians show narrower audiovisual temporal integration windows and quicker segregation, whereas experienced wine tasters seem more influenced by a wine's color in th...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1960361

Editorial: Emerging talents in human neuroscience: psychophysiology and cognitive neuroscience 2025

Gaia Risso 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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Simulation-based neural dynamics of stress-related negative cognitive bias in depressive symptoms: a dynamical EEG/ERP framework

Jianping Liu 2026-10-29 1 min 192 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis simulation-based computational study develops a non-linear state-space framework for stress-related negative cognitive bias and multivariate EEG/ERP observations.MethodsA three-state model represents negative bias, cognitive control, and arousal, while symptom load and normalized mo...

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Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

👤 Sivakumar Balasubramanian
📅 2026-10-20 📖 1 min 📝 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

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Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

👤 Lijuan Chen
📅 2026-10-29 📖 1 min 📝 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

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From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt
2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

👤 Mohammed Khaled Al-Hanawi 📅 2026-11-01 📖 1 min 📝 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

👤 Paul K Marsden 📅 2026-11-02 📖 1 min 📝 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Functional and gain-based vestibulo-ocular measures across serum 25(OH)D categories: a dissociation between fHIT performance and vHIT gain

👤 Alper Köycü 📅 2026-11-02 📖 1 min 📝 30 words LOW VISION

Summary: CONCLUSIONS AND SIGNIFICANCE: Low 25(OH)D was accompanied by reduced functional vestibulo-ocular performance in the vertical canal planes despite normal gains. fHIT may capture subclinical change that vHIT gain does not.

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

👤 Oluwatosin Oluwadare 📅 2026-11-02 📖 1 min 📝 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Exploring Spatial Inequities and Livability: A Mixed-Methods Study Using Artificial Intelligence and Community Insights

Rodrigo Siqueira Reis 2026-11-02 1 min 62 words LOW VISION

Summary: Rapid urbanization has intensified health and social inequities through the uneven distribution of infrastructure, services, and public space. In St. Louis, Missouri (USA), these disparities remain geographically concentrated and rooted in long-standing patterns of racial segregation and disinvestme...

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Association Between Quality of Life and Visual Function in Inherited Retinal Disease: A Cross-sectional Study in a Retinal Genetics Centre

Robert E MacLaren 2026-11-02 1 min 27 words LOW VISION

Summary: CONCLUSIONS: Microperimetry is a patient-relevant measure in inherited retinal diseases. This study demonstrates to stakeholders, such as regulatory bodies, how visual function measures translate to patients' quality-of-life.

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

Tiffany M Powell-Wiley 2026-11-02 1 min 70 words LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel
2026-10-25 1 min 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

Dongyu Wu 2026-10-30 1 min 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

Felix Blankenburg 2026-10-30 1 min 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964313/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42964381/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966015/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42966748/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261102223747&v=2.20.1

Coping strategies and dreams—The impact of waking-life distress on negative dream emotions.

2026-05-18 1 min 183 words DREAMING

Summary: The relationship between waking life and dreaming seems complex, and various studies show that dreams are influenced by waking-life experiences. However, only a few studies deal with the interactions between coping strategies in waking life and dream emotions. The main goal of this study is to inves...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000346

Tuesday Daily Thread: Advanced questions

👤 /u/ AutoModerator 📅 2026-11-03 📖 1 min 📝 288 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Wednesday Thread: Advanced Questions 🐍</h1> Dive deep into Python with our Advanced Questions thread! This space is reserved for questions about more advanced Python topics, frameworks, and best practices. <h2>How it Works:</h2> Ask A...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmdsr0/tuesday_daily_thread_advanced_questions/

🏔️ Garhwali Language Library v1.3.0 is now live on PyPI!

👤 /u/ National_Ad5713 📅 2026-11-03 📖 1 min 📝 150 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> An open-source Python toolkit bringing Garhwali language, linguistics and cultural heritage into the digital ecosystem. Key highlights: • 2.39M+ morphological word forms
 • 3,619+ lexical lemmas across 27 domains
 • 5 Garhwali dialects
 ...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmg1bw/garhwali_language_library_v130_is_now_live_on_pypi/

Pip 26.3 - Extended Pylock Support, Build Backend Warnings, and a caution on wheel installation

👤 /u/zurtex 📅 2026-11-03 📖 3 min 📝 614 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Today I got to be the release manager for pip 26.3! There are two big features, and one documentation update, that I think worth talking about: Extended Pylock Support You can now install extras and non-default groups from a pylock fil...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmgv01/pip_263_extended_pylock_support_build_backend/

SLAC researchers find proof of altermagnetism, a new magnetic phase

👤 gmays 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www6.slac.stanford.edu/news/2026-10-29-slac-researchers-find-proof-altermagnetism-new-magnetic-phase Comments URL: <a href="https://news.ycombinat...

🔗 **Read full article:**
https://www6.slac.stanford.edu/news/2026-10-29-slac-researchers-find-proof-altermagnetism-new-magnetic-phase

V100 SXM2 pinout and NVLink reference

👤 airhangerf15 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/pigeon-labs-stack/v100-sxm2-reference Comments URL: https://news.ycombinator.com/item?id=50344912 Points: 13 # Co...

🔗 **Read full article:**
https://github.com/pigeon-labs-stack/v100-sxm2-reference

The SEC Should Block Anthropic's IPO to Protect the Public

👤 cdrnsf 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.techpolicy.press/the-sec-should-block-anthropics-ipo-to-protect-the-public/ Comments URL: <a href="https://news.ycombinator.com/item?id=50344919">https://news.ycom...

🔗 **Read full article:**
https://www.techpolicy.press/the-sec-should-block-anthropics-ipo-to-protect-the-public/

WorkOS reports security breach

👤 werrett 📅 2026-11-02 📖 1 min 📝 217 words HACKER NEWS

> **Summary:** It's not on their website or Trust center yet, but we received the following breach notification from WorkOS: We are writing to notify you of a data security incident that we detected on October 30, 2026 and are continuing to investigate. We value your trust and want to keep you informed while ...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50345712

GUI64 – GUI for the Commodore 64 running on top of the built-in ROM

👤 ethanpil 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.c64-wiki.com/wiki/GUI64 Comments URL: https://news.ycombinator.com/item?id=50345735 Points: 12 # Comments: 1

🔗 **Read full article:**
https://www.c64-wiki.com/wiki/GUI64

Scaling Git at OpenAI

👤 makeramen 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://openai.com/index/scaling-git-at-openai/ Comments URL: https://news.ycombinator.com/item?id=50345924 Points: 56 # Comments: 11

🔗 **Read full article:**
https://openai.com/index/scaling-git-at-openai/

Intel stock collapse worsens as $3.5T chip rout spreads

👤 entryleveljobsc 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://finance.yahoo.com/markets/article/the-intel-stock-collapse-is-getting-worse--and-the-37-trillion-chip-rout-is-spreading-212154947.html">https://finance.yahoo.com/markets/article/the-intel-stock-collapse-is-getting-worse--and-the-37-trillion-chip-rout-is-spreading-212...

🔗 **Read full article:**
https://finance.yahoo.com/markets/article/the-intel-stock-collapse-is-getting-worse--and-the-37-trillion-chip-rout-is-spreading-212154947.html

Ask HN: Does anyone have a contact with Cleveland-area Ham club?

👤 stockresearcher 📅 2026-11-03 📖 1 min 📝 80 words HACKER NEWS

> **Summary:** I filled out the web form for the Cuyahoga Amateur Radio Club but it's probably been circular-filed... I gave some info about a large collection of radio gear that needs to be removed from a home or it will be put into a dumpster. A lot of it brand-new. If you know anybody at the club, please ...

🔗 **Read full article:**
https://news.ycombinator.com/item?id=50346164

DiscImageStudio: burn visible artwork onto CDs, DVDs and Blu-ray discs

👤 bpierre 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/JiaFeiMiao-K-Cat/DiscImageStudio Comments URL: https://news.ycombinator.com/item?id=50346849 Points: 59 # Comments: 16...

🔗 **Read full article:**
https://github.com/JiaFeiMiao-K-Cat/DiscImageStudio

Show HN: Hacker Times, a New York Times-Inspired Version Hacker News

👤 polpenn 📅 2026-11-03 📖 1 min 📝 33 words HACKER NEWS

> **Summary:** I thought it would be fun to browse HN as if it were a newspaper. The site updates once an hour. Enjoy! <hr /> Comments URL: https://news.ycombinator.com/item?id=50346981 Points: 61 # Comments: 19

🔗 **Read full article:**
https://hackertimes.ca/

Dogmatic Bayesianism Disorder

👤 noreply@blogger.com (Daniel Lakens) 📅 2025-12-06 📖 4 min 📝 910 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Note: This humorously intended sarcastic blog post directly mimics classifications of psychological disorders in the Diagnostic and Statistical Manual of the American Psychological Association, but Dogmatic Bayesianism is not in the DSM-5 as an actual psychologic...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/12/dogmatic-bayesianism-disorder.html

On the reliability and reproducibility of qualitative research

noreply@blogger.com (Daniel Lakens)
2026-02-08 14 min 2928 words

TWENTY PERCENT STATISTICIAN

Summary: <p class="MsoNormal">With my collaborators, I am increasingly performing qualitative research. I find qualitative research projects a useful way to improve my understanding of behaviors that I want to explore with other methods in the future. For example, some years ago I performe...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/02/on-reliability-and-reproducibility-of.html

New Papers: Optimal Filter Settings for ERP Research

Steve Luck
2024-02-04 2 min 569 words

ERP BOOT CAMP

Summary: <p class="">Zhang, G., Garrett, D. R., & Luck, S. J. (in press). Optimal filters for ERP research I: A general approach for selecting filter settings. Psychophysiology. https://doi.org/10.1111/psyp.14531 [<a href="https://www...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/2/4/optimal-filters

Q&A with Dr. Richard Carson, Professor of Biomedical Engineering and Radiology & Biomedical Imaging, Yale University and Yale School of Medicine

Adriel Carridice 2025-11-04 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-richard-carson-professor-of-biomedical-engineering-and-radiology-biomedical-imaging-yale-university-and-yale-school-of-medicine/

Disrupted imagery, disrupted scenes: mental scene construction in behavioural-variant frontotemporal dementia

1 min 18 words BRAIN RESEARCH

Summary: Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Mohamad El Haj, Guillaume Chapelet, Dimitrios Kapogiannis, Claire Boutoleau-Bretonnière

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600483X?dgcid=rss_sd_all

Impact of hearing loss on visual selective attention: Evidence from N2pc and resting-state networks

📖 1 min 📝 28 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Tongao Zeng, Xiaoou Bu, Han Zhu, Ziyao Liu, Hongli Xiao, Fuxiang Li, Wenjun Zhang, Fuyi Yang, Jie Chen, Hailin Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006361?dgcid=rss_sd_all

Resource competition controls the length-dependence of synaptic protein synthesis

👤 Rubisch, P., Raak, S. B., Wodke, J. A. H., Stefan, M. I., O'Donnell, C. 📅 2026-11-02 📖 1 min

📝 140 words BIORXIV NEUROSCIENCE

Summary: Synaptic plasticity is dependent on the availability of specific proteins at the synapse with distinct requirements for potentiation and depression. Specifically, experimental evidence suggests an upregulation of short proteins and a downregulation for long proteins during long-term depression (LTD)...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763010v1?rss=1

In-vivo MRI of noradrenergic and dopaminergic nuclei predicts future cognition and Alzheimers disease progression

Zhao, W., Guzman, S. J., Yoo, H. J., Ruiz, K., Chen, K., Moheban, A., Nation, D. A., Dahl, M. J., Mather, M.

2026-11-02 1 min 162 words BIORXIV NEUROSCIENCE

Summary: Noradrenergic and dopaminergic nuclei modulate cognition, and post-mortem data indicate they are affected early in the neurodegenerative cascade of Alzheimers disease. We derived in-vivo locus coeruleus (LC) and substantia nigra-ventral tegmental area (SN-VTA) MRI integrity measures from 739 ADNI pa...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.29.762942v1?rss=1

Assessing reinforcement learning and vigor with an online decision-making battery

Wenke, S., Granwald, T., Ferrante, A., Nilsson, M., Ekström, I., Papenberg, G., Guitart-Masip, M.

2026-11-02 1 min 191 words BIORXIV NEUROSCIENCE

Summary: Reinforcement learning and vigor are two fundamental aspects of decision-making modulated by dopamine. Although vigor is highly relevant for mood disorders, it has been overlooked by previous batteries characterizing decision-making abilities. We developed a test battery with two RL tasks (RL/workin...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763028v1?rss=1

Natural predator-associated chemosignals evoke distinct spatiotemporal patterns of risk assessment and vomeronasal population responses

Wang, J., Haran, V., Meeks, J. P.

2026-11-02 1 min 208 words BIORXIV NEUROSCIENCE

Summary: Natural predator-associated chemosignals elicit diverse defensive behaviors, yet it remains unclear how complex natural cues shape the spatiotemporal organization of risk assessment and activate vomeronasal sensory neurons (VSNs) and downstream circuits in the mammalian accessory olfactory system. H...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.763133v1?rss=1

Sex differences in fear extinction are conditional, not categorical

Nguyen, M. T., Didachos, A., Roozendaal, B., Henckens, M. J. A. G., Gulmez Karaca, K.

2026-11-02 1 min 233 words BIORXIV NEUROSCIENCE

Summary: Threat memories are dynamic, and their behavioral expression can change with subsequent experience and the passage of time. Extinction training can reduce conditioned fear responses, yet these responses may re-emerge over time through spontaneous recovery. Although biological sex has been implicated...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.762318v1?rss=1

Feedback reliability modulates learning-related dynamics of midfrontal error monitoring

Gawlowska, M.

2026-11-02 1 min 153 words BIORXIV NEUROSCIENCE

Summary: Learning from feedback requires the brain to evaluate both performance outcomes and the reliability of the information used for learning. This study examined the dynamics of neural performance monitoring during deterministic (100% reliable) and probabilistic (75% reliable) associative learning using...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.762838v1?rss=1

Endothelin-1 from Tongue Glands Gates 1 Sodium Taste and Intake

Chen, L.-T., Xia, X., Soto, E. R., Ostroff, G. R., Zhang, J.

2026-11-02 1 min 181 words

BIORXIV NEUROSCIENCE

Summary: Sodium is an essential nutrient. To survive sodium deficiency, animals dramatically increase their appetite for salt. However, whether and how sodium depletion impacts the taste of salt to drive sodium intake is unknown. Here we show that tongue gland-derived endothelin-1 (EDN1) regulates the amilor...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762212v1?rss=1

Mapping Endogenous Psychedelic Chemistry: Indolethylamine N-Methyltransferase (INMT) Localizes Proximal to the Golgi Apparatus of Human Neurons

Will, K., Lazaro Martinez, S., Reis, S. A., Haggarty, S. J.

2026-11-02 1 min 211 words

BIORXIV NEUROSCIENCE

Summary: N,N-dimethyltryptamine (DMT) is a serotonergic psychedelic with a history of ethnobotanical use and emerging clinical efficacy for neuropsychiatric disorders. Following Julius Axelrod's pioneering studies showing DMT production in human tissues, DMT has been proposed as an endogenous neuromodulator ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.758666v1?rss=1

LAMA2-related muscular dystrophy alters Schwann cell calcium homeostasis and increases peripheral nerve inflammation

👤 Wang, J., Khati, M., Heron, M., Munir, A., Akinyele, O., Johnson, M. A., Vockley, J., Pandey, U. B., Kemaladewi, D. U.

📅 2026-11-02 📖 1 min 📝 171 words BIORXIV NEUROSCIENCE

Summary: LAMA2-related muscular dystrophy (LAMA2-RD) affects skeletal muscle and peripheral nerves, yet the consequences of LAMA2 deficiency on peripheral nerve integrity beyond early postnatal development remain incompletely understood. We investigated sciatic nerve architecture and molecular alterations ac...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762325v1?rss=1

Nucleus accumbens acetylcholine dips track reward value and temporal costs to shape delay discounting

Cavallaro, J., Zuelke, D. R., Maysonet, C. M., Salazar, R. E., Setara, R., Gallo, E. F.

2026-11-02

1 min

186 words

BIORXIV NEUROSCIENCE

Summary: As an organism chooses between competing rewards, it must often integrate learned reward value with the changing costs associated with obtaining them. Yet, the brain mechanisms through which these factors jointly guide choice are not well understood. Previous studies have shown that dopamine signals...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.760841v1?rss=1

Anterior cingulate neurons perform parallel processing of fixated and peripheral information to guide information sampling and choice

2026-11-02

1 min

0 words

NATURE NEUROSCIENCE SUBJECTS

Read full article:
https://www.nature.com/articles/s41467-026-78945-4

Human subthalamic neurons track speech features and couple with auditory cortex during preparatory listening

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78957-0

iTBS and cTBS to an individualized dlPFC target produce differential working memory improvement and session-dependent putamen deactivation

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11295-5

A brain of many microglia origins

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02509-4

Misfolded tau propagates via templated seeding

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02512-9

Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02530-7

It’s time to split

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02511-w

Sleep-sensitive neurons mediate consolidation

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41583-026-01101-w

It’s time to split

👤 William P. Olson 📅 2026-11-02 📖 1 min 📝 12 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02511-w It’s time to split

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02511-w

A brain of many microglia origins

Ioana A. Marin 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02509-4 A brain of many microglia origins

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02509-4

Correcting automatic decisions

Luis A. Mejia 2026-11-02 1 min 11 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02510-x Correcting automatic decisions

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02510-x

Misfolded tau propagates via templated seeding

Qingzhong Ren 2026-11-02 1 min 14 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02512-9 Misfolded tau propagates via templated seeding

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02512-9

Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

Michael W. Cole 2026-11-02 1 min 22 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02530-7 Publisher Correction: Network geometry shapes the transformation of task representations across the human cortex

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02530-7

From pose estimation accuracy to motor biomarker reliability: a reliability-centred evaluation framework for markerless movement analysis

Christine Fernandez-Maloigne
2026-10-01 1 min 246 words

FRONTIERS NEUROINFORMATICS

Summary: Markerless pose estimation has become an essential tool for quantitative movement analysis in neuroscience. However, current evaluation frameworks primarily rely on frame-wise spatial accuracy metrics, which provide only limited information about the reliability of the downstream kinematic biomarker...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1790241

Altered interictal alpha-band organization in childhood absence epilepsy

Justyna Paprocka
2026-10-23 1 min 318 words

FRONTIERS NEUROINFORMATICS

Summary: ObjectiveInterictal background EEG in childhood absence epilepsy (CAE) may appear normal. We asked whether drug-naive children with CAE differ from controls in alpha frequency and the regional distribution of phase-lagged synchronization, and how these differences depend on participant-specific surr...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1875377

Short-term synaptic plasticity modulates the outcome of neurodegenerative diseases

Chi Chung Alan Fung 2026-10-23 1 min 264 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAlzheimer's disease (AD) is characterized by progressive neuronal loss and synaptic dysfunction that disrupt memory-encoding neural circuits. Short-term synaptic plasticity (STP), which governs transient activity-dependent changes in synaptic efficacy on millisecond-to-second timescales,...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1923602

Orientation detection for self-calibrating neuromorphic systems

Elisabetta Chicca 2026-10-23 1 min 183 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Bioinspired sensory architectures help build adaptive touch systems with efficient hardware implementations. Orientation detection is a fundamental capability for manipulating objects in real-world scenarios, with diverse applications in robotic manipulation and manufacturing assessment processes. I...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1878557

A systems-level energetic framework for conscious processing: energetic organization as a constraint on large-scale neural integration

Holly Pollard-Wright 2026-10-23 1 min 259 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Conscious processing is associated with coordinated interactions among distributed neural systems, yet the biological constraints that enable and sustain large-scale neural integration remain incompletely understood. Contemporary theories emphasize computational, informational, and network mechanism...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1966072

Immediate effect of gait training using the hybrid assistive limb in patients in the acute phase of stroke: a prospective pilot study

Yasushi Takagi 2026-10-23 1 min 360 words FRONTIERS NEUROROBOTICS

Summary: IntroductionThe effectiveness of Hybrid Assistive Limb (HAL) gait training in patients in the acute phase of stroke remains unclear. We prospectively assessed the immediate effects of HAL gait training in this patient population.MethodsIn this prospective pilot study, four patients with acute-phase ...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1985500

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261102172642&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

👤 Arnaud Delorme 📅 2026-08-01 📖 1 min 📝 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

Chuxu Wang 2026-10-31 1 min 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

N Chaitanya Kumar 2026-10-31 1 min 68 words LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

Mohammed Khaled Al-Hanawi 2026-11-01 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

👤 Paul K Marsden 📅 2026-11-02 📖 1 min 📝 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Functional and gain-based vestibulo-ocular measures across serum 25(OH)D categories: a dissociation between fHIT performance and vHIT gain

👤 Alper Köycü 📅 2026-11-02 📖 1 min 📝 30 words LOW VISION

Summary: CONCLUSIONS AND SIGNIFICANCE: Low 25(OH)D was accompanied by reduced functional vestibulo-ocular performance in the vertical canal planes despite normal gains. fHIT may capture subclinical change that vHIT gain does not.

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

Oluwatosin Oluwadare 2026-11-02 1 min 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel
2026-10-25 1 min 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

👤 Vanja Garaj 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

👤 Jarred Yasuhara-Bell 📅 2026-10-27 📖 1 min 📝 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

Felix Blankenburg 2026-10-30 1 min 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang 📅 2026-10-30 📖 1 min 📝 64 words TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

Chuhuai Wang
2026-10-31 1 min 47 words TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

Yulong Bai 2026-10-30 1 min 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

Teri Kim 2026-10-30 1 min 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 69 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu | 2026-11-01 | 1 min | 46 words | BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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CPython core dev Larry Hastings on testing race conditions in free-threaded Python and AI-coding

/u/miabajic | 2026-11-02 | 1 min | 239 words | REDDIT PYTHON

Summary: <!-- SC_OFF --> <p>Hey everyone! In my free time I host Behind the Commit, a podcast with open-source maintainers. I wanted to talk to Larry Hastings because free-threaded Python brought lots of challenges with testing race conditions -- I myself had no idea how to test code without G...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xm18lx/cpython_core_dev_larry_hastings_on_testing_race/

FastAPI Cloud is now generally available

👤 /u/tiangolo 📅 2026-11-02 📖 1 min 📝 68 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I'm excited to share with you all that FastAPI Cloud is now generally available: https://fastapicloud.com/blog/fastapi-cloud-generally-available/ <hr /> Note: I thought it could be...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xm7oif/fastapi_cloud_is_now_generally_available/

Skill to Create Product Videos

👤 aaitor 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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Sneaking into the auditorium for the Windows 2000 launch rehearsal

👤 ibobev 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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Astra can read and write weights to improve Qwen

👤 dbreunig 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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Agentic backdoors via pretraining poisoning

👤 gmays 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2610.37741 Comments URL: https://news.ycombinator.com/item?id=50341334 Points: 71 # Comments: 19

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https://arxiv.org/abs/2610.37741

Ubuntu still has not released patch for Linux 0day RCE in almost 3 weeks

👤 majorchord 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://freenode.net/article/ubuntu-lts-kernels-still-unfixed-for-critical-cve-2026-98432 Comments URL: <a href="https://news.ycombinator.com/item?id=50341462">https://news....

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https://freenode.net/article/ubuntu-lts-kernels-still-unfixed-for-critical-cve-2026-98432

Ballot Rejections Have Soared in Rural Counties as USPS Reduces Mail Service

👤 ck2 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.propublica.org/article/mail-ballot-delays-postal-service-midterms Comments URL: <a href="https://news.ycombinator.com/item?id=50341616">https://news.ycombinator.com/item?id=...

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https://www.propublica.org/article/mail-ballot-delays-postal-service-midterms

Egoless Mathematics in the AI Era

👤 smilelamp 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://terrytao.wordpress.com/2026/11/02/egoless-mathematics-in-the-ai-era/ Comments URL: <a href="https://news.ycombinator.com/item?id=50341736">https://news.ycombinator.com/item?id=50...

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https://terrytao.wordpress.com/2026/11/02/egoless-mathematics-in-the-ai-era/

Oracle files to sell $20B in new stock, bonds to finance AI buildout

👤 smartmic 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.cnbc.com/2026/11/02/oracle-files-to-sell-20-billion-in-new-stock-bonds-to-finance-ai-buildout.html Comments URL: <a href="https://news.ycomb...

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https://www.cnbc.com/2026/11/02/oracle-files-to-sell-20-billion-in-new-stock-bonds-to-finance-ai-buildout.html

BYD sold more EVs last month than the US did all quarter, topping 300,000

👤 cisc 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://electrek.co/2026/11/02/byd-sold-more-evs-last-month-than-the-us-did-all-quarter/ Comments URL: <a href="https://news.ycombinator.com/item?id=50344181">https://news.yc...

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https://electrek.co/2026/11/02/byd-sold-more-evs-last-month-than-the-us-did-all-quarter/

Can't fish Ethernet? Use "invisible" fiber instead

👤 _mpn_ 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://momandpop.network/2026/11/02/cant-fish-ethernet-use-invisible-fiber-instead/ Comments URL: <a href="https://news.ycombinator.com/item?id=50344222">https://news.ycombinato...

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https://momandpop.network/2026/11/02/cant-fish-ethernet-use-invisible-fiber-instead/

There are 13 ways to analyse a replication study, but only one of them is coherent.

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-03 📖 6 min 📝 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice 2026-09-23 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

🔗 **Read full article:**
https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

A prospective randomized study of patient-centered AI for clinical visit preparation

👤 Xincheng Tan, Matthew Thompson, Xiang Ji, Angelica Willis, Will Vaughan, Roma Ruparel, Michael A. Sanchez, Sean Chang, Joseph Xu, Bhavna Daryani, Preeti Singh, Pinal Bavishi, Kejia Chen, Fan Zhang, Priya Gupta, Ernest Lim, Christina Chen, Bob Lou, Dimitrios Antos, Zach Wasson, Ray Luo, Fadi Yousif, Florence Thng, Mark Malhotra, Joseph Breda, Daniel McDuff, Kavi Goel, Katherine Chou, Dale R. Webster, Avinatan Hassidim, Yossi Matias, Po-Hsuan Cameron Chen, Yun Liu, Mike Schaekermann, Quang Duong, Jackie M. Barr

📅 2026-11-02 📖 1 min 📝 282 words ARXIV CS HC

Summary: arXiv:2610.41437v1 Announce Type: new Abstract: Patients often arrive at brief, time-constrained clinical visits unsure of what to expect or ask, and millions of adults now turn to general-purpose AI chatbots to prepare beforehand. Yet whether conversational AI improves visit readiness remains uncl...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41437

Hybrid Computer Use: A Unified Programmatic, Semantic, and Visual Interface for General Agents

Haoqing Wang, Xiaoyi Zhang, Sicheng Fan, Zhexin Zhang, Xinjie Zhang, Jiahao Li, Bin Li, Yan Lu

2026-11-02 1 min 212 words ARXIV CS HC

Summary: arXiv:2610.41432v1 Announce Type: new Abstract: Computer-use agents increasingly tackle tasks ranging from everyday desktop activities to long-horizon professional workflows. Yet interfaces centered on screenshots and mouse-and-keyboard actions can impose a human-interaction prior, requiring repeat...

Read full article:
https://arxiv.org/abs/2610.41432

When History Overrides Intent: Relational Schema Lock-In in Human-LLM Interaction

Wei Yang

2026-11-02 1 min 148 words ARXIV CS HC

Summary: arXiv:2610.41367v1 Announce Type: new Abstract: Long-context interaction and persistent memory allow LLM assistants to develop history-grounded understandings of users. When users change direction, however, an understanding built through earlier exchanges may conflict with their current instruction...

Read full article:
https://arxiv.org/abs/2610.41367

Toward Low-Burden Image-Based Dietary Assessment Through Human-AI Collaborative Correction

Darren Liu, Andrew Lu, Zhiyuan "Ryan" Wang, Hanqi Luo, Xiao Hu, Runze Yan

2026-11-02

1 min

152 words

ARXIV CS HC

Summary: arXiv:2610.41300v1 Announce Type: new Abstract: Image-based dietary assessment can automate initial food logging, but meal photographs often leave food identity, hidden ingredients, preparation, and portion uncertain, making correction a recurring part of use. We introduce Human-AI Collaborative Co...

Read full article:
https://arxiv.org/abs/2610.41300

Imputing Missing Data in Longitudinal Life-Logging Athlete Data

Saaz Kaur Sahdra, Kalina Yacef, HaeJin Lee, Judy Kay

2026-11-02

1 min

187 words

ARXIV CS HC

Summary: arXiv:2610.41226v1 Announce Type: new Abstract: Long-term, diverse personal informatics data enables athletes to understand and act on their health and performance goals, whether assessing readiness right now, tracking how soreness relates to training load over time, or predicting how sleep affects...

Read full article:
https://arxiv.org/abs/2610.41226

Flash Word : ESL Listening Support with Subtitle Skimming

👤 Yuzu Saijo, Jun Rekimoto
📅 2026-11-02 📖 1 min 📝 131 words ARXIV CS HC

Summary: arXiv:2610.41098v1 Announce Type: new Abstract: Listening to English audio with English subtitles is thought to be effective for ESL students to improve listening comprehension skills. However, for those who learn English as a second language it can be hard to finish English subtitles. Also they ar...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41098

Out of sync: Accommodation, evaluation, and emergence in five weeks with an agentic AI teammate

👤 Nia Nixon, Jaeyoon Choi, Pedro Martins de Bastos, Mohammad Amin Samadi
📅 2026-11-02

📖 1 min 📝 216 words ARXIV CS HC

Summary: arXiv:2610.41057v1 Announce Type: new Abstract: As agentic AI in education shifts from reactive tools toward partners that take initiative and persist over time, a central question is how learners relate to, evaluate, and benefit from artificial agency. We conducted a longitudinal experiment in an ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41057

Measuring Personal Information Exposure in Mental Health-Related Conversations with LLMs

Synthia Qia Wang, Ranya Sharma, Christine Lee, Sarah Keedy, Marshini Chetty, Nick Feamster

2026-11-02 1 min 155 words ARXIV CS HC

Summary: arXiv:2610.41045v1 Announce Type: new Abstract: Large language models (LLMs) are increasingly used for informal mental health support, yet the privacy implications of conversational interaction remain poorly understood. Prior LLM privacy research has focused largely on model-side risks such as memo...

Read full article:
https://arxiv.org/abs/2610.41045

Robot Navigation and Canine Welfare: A Pilot Study Surfacing Challenges for Cross-Species Legibility and Proxy Validity in Animal-Machine Interaction

Haoxiang Zhang, Nancy Smith

2026-11-02 1 min 159 words ARXIV CS HC

Summary: arXiv:2610.40921v1 Announce Type: new Abstract: Autonomous delivery robots increasingly share residential spaces with companion animals, yet navigation behavior is rarely designed with animal welfare in mind. This paper presents a pilot study evaluating a curve-out maneuver, a laterally arcing pass...

Read full article:
https://arxiv.org/abs/2610.40921

Personalised 3D Mnemonics for Vocabulary Learning in Context

👤 Karolina Trajkovska, Matja\v{z} Kljun, Maheshya Weerasinghe, Klen \v{C}opi\v{c} Pucihar

📅 2026-11-02 📖 1 min 📝 185 words ARXIV CS HC

Summary: arXiv:2610.40760v1 Announce Type: new Abstract: Augmented reality (AR) can support vocabulary learning by placing mnemonic cues in the physical contexts where new words are encountered. Prior work has augmented real objects with bilingual labels, audio, keywords, and 3D visualisations of those keyw...

🔗 **Read full article:**
https://arxiv.org/abs/2610.40760

Critical Flicker Fusion Frequency As An Experience-Restricted Constraint On Visual Temporal Resolution: What Does And Does Not Change It

Natalia D. Rydzenska, Pawel J. Winklewski, Michal W. Blaszczyk-Niezgoda, Anna B. Marcinkowska

2026-11-02 1 min 250 words ARXIV QBIO NC

Summary: arXiv:2607.29068v3 Announce Type: replace Abstract: Experience-dependent plasticity is fundamental to adaptive behaviour, yet the conditions under which basic sensory timing can be modified in adulthood remain poorly specified. Critical flicker fusion frequency (CFFF), the threshold at which flicke...

Read full article:
https://arxiv.org/abs/2607.29068

Perceived vertical and eye level as one orientation process: a closed-form account of the Li-Matin combination rules for egocentric space

A. Y. Shavit

2026-11-02 1 min 302 words ARXIV QBIO NC

Summary: arXiv:2607.19681v5 Announce Type: replace Abstract: Li and Matin (2005a, 2005b) reported three regularities of the induced visually-perceived vertical (VPV), the direction judged upright. For small tilts, a peripheral line shifts it nearly linearly with tilt, two lines combine nearly linearly, and ...

Read full article:
https://arxiv.org/abs/2607.19681

Message interaction dynamics covary with brain volume across mammalian connectomes

Yan Hao, Hannah Lax, Marc-Thorsten H\"utt, Daniel J. Graham

2026-11-02 1 min

197 words

ARXIV QBIO NC

Summary: arXiv:2505.15477v3 Announce Type: replace Abstract: Brain network communication models typically assume that signals propagate independently, despite the high network density and small diameter of mammalian connectomes, where interactions among simultaneously propagating messages are likely. We inv...

Read full article:
https://arxiv.org/abs/2505.15477

Building functional and mechanistic models of cortical computation based on canonical cell type connectivity

Arno Granier, Katharina A Wilmes, Mihai A Petrovici, Walter Senn

2026-11-02 1 min

154 words

ARXIV QBIO NC

Summary: arXiv:2504.03031v3 Announce Type: replace Abstract: Neuronal circuits of the cerebral cortex are the structural basis of mammalian cognition. The same qualitative components and connectivity motifs are repeated across functionally specialized cortical areas and mammalian species, suggesting a singl...

Read full article:
https://arxiv.org/abs/2504.03031

Lattice-Based Pruning of Recurrent Neural Networks via Poset Modeling: A Corrected Theory and a Matched-Sparsity Evaluation

Rakesh Sengupta

2026-11-02 1 min 197 words ARXIV QBIO NC

Summary: arXiv:2502.16525v2 Announce Type: replace Abstract: Magnitude pruning of recurrent neural networks (RNNs) ignores how information flows between neurons. An earlier version proposed to model RNN dependencies as a partially ordered set (poset) and to retain the meet-irreducible elements of its depend...

🔗 **Read full article:**
https://arxiv.org/abs/2502.16525

Explainable AI-Driven Neural Activity Analysis in Parkinsonian Rats under Electrical Stimulation

Jibum Kim, Hanseul Choi, Gaeun Kim, Sunggu Yang, Eunha Baeg, Donggue Kim, Seongwon Jin, Sangwon Byun

2026-11-02 1 min 167 words ARXIV QBIO NC

Summary: arXiv:2502.12471v2 Announce Type: replace Abstract: Parkinson's disease (PD) is characterized by motor dysfunction and abnormal neural oscillations. This study combined graphene-based cortical stimulation with an explainable artificial intelligence (XAI) framework to investigate spatial and frequen...

🔗 **Read full article:**
https://arxiv.org/abs/2502.12471

Hyperbolic embedding of higher-order simplicial networks

Martin Guillemaud, Vito Latora, Mario Chavez

2026-11-02 1 min 177 words

ARXIV QBIO NC

Summary: arXiv:2610.41684v1 Announce Type: new Abstract: Many biological interactions involve groups rather than pairs, as when several proteins assemble into a multi-protein complex. Simplicial complexes naturally capture these collective interactions. Existing hyperbolic embeddings represent network hiera...

🔗 **Read full article:**
https://arxiv.org/abs/2610.41684

Expression of White and Gray Matter Autism Functional Network Connectivity Signatures in Infancy

Qiang Li, Jingyu Liu, Zening Fu, Armin Iraji, Sarah Shultz, Vince D. Calhoun

2026-11-02 1 min 211 words

ARXIV QBIO NC

Summary: arXiv:2610.40839v1 Announce Type: new Abstract: Functional connectivity studies have provided valuable insights into altered brain organization in autism spectrum disorder (ASD). However, most functional connectivity studies focus exclusively on either gray matter (GM) regions or white matter (WM) ...

🔗 **Read full article:**
https://arxiv.org/abs/2610.40839

Prolonged visual-motor dual-task effects on dynamic stability in children with ADHD subtypes

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https://www.nature.com/articles/s41598-026-77122-3

DIA proteomics identifies time-of-day-dependent hippocampal proteomic signatures associated with exercise timing in type 2 diabetic mice

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https://www.nature.com/articles/s41598-026-76605-7

Factors influencing plasma brain-derived neurotrophic factor (BDNF) in adolescent sport participants

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https://www.nature.com/articles/s41598-026-76457-1

Association of diffusion along the perivascular spaces with clinical symptoms and dopaminergic degeneration in Parkinson’s disease

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🔗 **Read full article:**
https://www.nature.com/articles/s41531-026-01620-4

Genomics for chronic overlapping pain conditions

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🔗 **Read full article:**
https://www.nature.com/articles/s41576-026-01029-y

A multi-institutional investigation of psilocybin’s effects on mouse behavior

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https://www.nature.com/articles/s41593-026-02500-z

State space models naturally produce time cell and oscillatory behaviors and scale to abstract cognitive functions

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https://www.nature.com/articles/s41467-026-79286-y

Tissue-specific lipidomic alterations in aging rats

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https://www.nature.com/articles/s41598-026-77040-4

Microglia-dependent PDGFB regulation of neuronal KCND2 in the hypothalamic paraventricular nucleus determines propofol anesthesia induction

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🔗 **Read full article:**
https://www.nature.com/articles/s41467-026-78798-x

Author Correction: mTORC1 drives cell-autonomous astrocyte reactivity in tuberous sclerosis

📅 2026-11-02 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11350-5

Serotonin signaling enhances phagocytosis by brain macrophages through Htr2b and alleviates brain injury after stroke

👤 Peiying Li 📅 2026-10-28 📖 1 min 📝 45 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 28 October 2026; doi:10.1038/s41593-026-02482-y The serotonin receptor Htr2b activation enhances lysosome biogenesis and phagocytic clearance of debris to mitigate neuroinflammation and brai...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02482-y

A multi-institutional investigation of psilocybin’s effects on mouse behavior

Robert C. Malenka
2026-11-02 1 min 44 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 02 November 2026; doi:10.1038/s41593-026-02500-z In a study across five independent labs, 2 mg kg−1 psilocybin in mice acutely increased avoidance-related behaviors in a robust, reproducible...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02500-z

Bite-block effects on native s/ʃ and English r/l in child speakers of a language with/without this contrast

F. Sayako Earle
2026-11-02 1 min 171 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Learning to produce non-native speech contrasts requires developing new phonological categories as well as new articulatory-perceptual mappings for these new categories. However, within the same speakers, it remains unclear how these new categories compare to contrasts that are already fully establi...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949253

From biomechanical data to robotic execution: design and control of an intelligent Tuina system

Wei Yao 2026-11-02 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: IntroductionTraditional Chinese medicine (TCM) Tuina (massage) relies heavily on practitioners' physical stamina and subjective experience, leading to operational variability that hinders therapeutic standardization and scalable rehabilitation service delivery. To address this limitation, this study...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1937295

Mixed-timestep spiking neural networks: a temporal alignment framework for high-efficiency neuromorphic computing

Liang Chen 2026-11-02 1 min 229 words FRONTIERS NEUROSCIENCE

Summary: Spiking Neural Networks (SNNs) are energy-efficient due to their event-driven computation and sparse spike-based communication. However, most existing SNN models adopt a uniform time-step across all layers, which limits temporal flexibility and leads to suboptimal accuracy latency trade-offs under l...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1783326

Evidence for decoder-label-regime-dependent effects of complementary leaky integrate-and-fire temporal-state decoding in motor-imagery EEG

Dong Zhang 2026-11-02 1 min 272 words FRONTIERS NEUROSCIENCE

Summary: BackgroundMotor-imagery electroencephalography often provides limited subject-specific supervision. This study tested whether complementary leaky integrate-and-fire (CLIF)-family temporal-state decoding shows a decoder-level advantage after the EEG representation is fixed, how that advantage changes...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1966366

Diurnal cortisol, perceived sleep quality, actigraphic sleep and depression: an exploratory pilot in patients with head and neck cancer

Sandra E. Sephton 2026-11-02 1 min 231 words FRONTIERS NEUROSCIENCE

Summary: BackgroundIn patients with head and neck cancer (HNC), perceived sleep quality, actigraphy-derived sleep parameters, depressive symptoms, and diurnal cortisol may represent related behavioral and biological processes. This exploratory study tested a priori associations among these constructs.Methods...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1972777

Case Report: Clinical improvement of localized chronic eczema following multisegmental paraspinal electroacupuncture

Jianhua Tang 2026-11-02 1 min 172 words FRONTIERS NEUROSCIENCE

Summary: BackgroundNeuroimmune interactions provide a rationale for studying somatosensory neuromodulation in inflammatory disorders, but a single clinical observation cannot establish an autonomic mechanism. We report a case of localized chronic eczema that improved after multisegmental paraspinal electroac...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnins.2026.1905399

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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https://pubmed.ncbi.nlm.nih.gov/39786979/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261102081041&v=2.20.1

NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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https://pubmed.ncbi.nlm.nih.gov/39870674/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261102081041&v=2.20.1

Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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https://pubmed.ncbi.nlm.nih.gov/40568426/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261102081041&v=2.20.1

Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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https://pubmed.ncbi.nlm.nih.gov/40800958/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261102081041&v=2.20.1

Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

> **Summary:** The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

> **Summary:** The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Intramedullary Spinal Tuberculosis Mimicking Neoplasm with Ocular Involvement in an Adolescent: A Case Report

Indri Febriani 2026-10-31 1 min 72 words LOW VISION

Summary: Tuberculosis (TB) continues to represent one of the key health problems of high-burden nations. The childhood population is particularly vulnerable to extrapulmonary TB. Spinal intramedullary TB is extremely uncommon and can imitate neoplastic conditions, leading to a late diagnosis with fatal outco...

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Framework-informed analysis of practice: A qualitative study on climate resilience in primary care in Austria

👤 Katharina Brugger 📅 2026-10-31 📖 1 min 📝 59 words LOW VISION

Summary: CONCLUSIONS: PCPs play a vital role in local climate resilience but require tailored, context-specific guidance aligned with PC's integrated, small-scale, multiprofessional nature. Strategic macro-level objectives are required to support low-barrier implementation of resilience measures. Prioritizin...

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Early applications of apple vision pro in clinical medicine and medical education: a scoping review

👤 Chuxu Wang 📅 2026-10-31 📖 1 min 📝 58 words LOW VISION

Summary: The Apple Vision Pro (AVP), a next-generation spatial computing device, has demonstrated potential applications in multidisciplinary clinical diagnosis, treatment, and medical education, by leveraging its unique technical architecture and performance advantages. In clinical practice, its fundamental...

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A Distributed Vision Transformer-Based Deep Learning Framework for Early Detection and Classification of Lumbar Disc Degeneration

👤 N Chaitanya Kumar
📅 2026-10-31 📖 1 min 📝 68 words

LOW VISION

Summary: Lumbar disc degeneration (LDD) is a principal structural cause of chronic low back pain. Magnetic resonance imaging (MRI) is the reference modality for assessing the intervertebral discs, but severity grading is performed manually, is limited by inter-observer variability, and becomes a bottleneck a...

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A Result-Auditable Protocol for Low-Data Classification of Blue-and-White Porcelain Botanical Motifs Using Frozen Chinese-CLIP Features

👤 Chenghao Yu
📅 2026-10-31 📖 1 min 📝 58 words

LOW VISION

Summary: Low-data visual classification is common in cultural heritage documentation, where expert labels are expensive, and image sources are heterogeneous. This protocol provides result-level audit support for classifying five botanical motifs-peony, lotus, chrysanthemum, plum blossom, and bamboo-on blue-a...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

Mohammed Khaled Al-Hanawi 2026-11-01 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

Paul K Marsden 2026-11-02 1 min 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel
2026-10-25 1 min 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Improvement in naming with tDCS and alteration of functional connectivity in post-stroke aphasia: a randomized controlled trial

👤 Dongyu Wu 📅 2026-10-30 📖 1 min 📝 58 words TDCS TACS TRNS

Summary: CONCLUSIONS: Anodal tDCS over the left SPT can improve picture naming and auditory comprehension in subacute post-stroke aphasia. Potential mechanisms include enhanced connectivity within the left-hemisphere dorsal/ventral streams, the executive control network, and between the bilateral temporal lo...

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Parietal tDCS and tactospatial working memory: Evidence for a behavioral benefit in a sham-controlled crossover study

Felix Blankenburg 2026-10-30 1 min 60 words TDCS TACS TRNS

Summary: The posterior parietal cortex (PPC) contributes to spatial working memory (WM), but whether parietal transcranial direct current stimulation (tDCS) can modulate tactospatial WM remains unclear. Thirty-two adults completed a double-blind, sham-controlled, within-subject crossover study with two count...

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Effect of Electrical Stimulation on Remyelination in Preclinical Models of Multiple Sclerosis: A Systematic Review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words

TDCS TACS TRNS

Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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Prefrontal circuitry abnormalities and cognitive impairment in adolescents with early-onset psychosis

Jaymin Upadhyay 2026-10-29 1 min 62 words FNIRS

Summary: Early-onset psychosis (EOP) is associated with marked cognitive impairment and poorer health-related and psychosocial outcomes compared to adult-onset illness, yet the neural mechanisms underlying these deficits during adolescence remain incompletely understood. Despite evidence implicating dorsolat...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

Tobias Grossmann 2026-10-29 1 min 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

Chen-Hui Pei 2026-10-30 1 min 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

Dandan Wan 2026-10-31 1 min 17 words FNIRS

Summary: CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: greater unpredictability in infancy, local reorganization in childhood

Kanghui Li 2026-10-31 1 min 69 words FNIRS

Summary: The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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Differential effects of γ-state-triggered thalamic stimulation on mechanical sensitivity in neuropathic rats

👤 Shouyan Wang 📅 2026-10-30 📖 1 min 📝 36 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The tested WP-defined γ-state-triggered CL protocols were associated with differing changes in mechanical sensitivity in neuropathic rats. These findings support further evaluation of this approach as a candidate framework for adaptive neuromodulation of neuropathic pain.

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EEG Feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

👤 Aishi Zhou 📅 2026-10-30 📖 1 min 📝 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Implicit learning of melodic structure: A role for pitch?

2024-01-22 1 min 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

> **Summary:** I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

From Charlemagne to the Kitchen Boy: Reading the Hamburg Rathaus

👤 mertbio 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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👤 mw888 📅 2026-11-02 📖 1 min 📝 17 words HACKER NEWS

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China's First Frontier AI Export Controls: Mofcom Announcement No. 48

👤 jonnybgood 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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BioWare co-founder: game execs "have never made a game" and don't get it

👤 entryleveljobsc 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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Show HN: I've updated my Linux System Call website and API

👤 guitmz 📅 2026-11-02 📖 1 min 📝 57 words HACKER NEWS

> **Summary:** <p>I first released those in back in 2020 and now I finally added more architectures, Kernel selector, syscall descriptions and a few other niceties. It should work a bit better on mobile devices too but I am no CSS expert.<p>The API is at <a href="https://api.syscall.sh" rel="nofollow">https://api....

🔗 **Read full article:**
https://syscall.sh/

The future in old computer magazines (1984–2004)

👤 tudell 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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Supermarket freezers in Belgium and Sweden hit by suspected cyber-attack

👤 xqcgrek2 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

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https://www.bbc.com/news/articles/cq1kd1nee80eo

Supermarket freezers across Europe hit by second suspected cyber-attack

👤 fergie 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.theguardian.com/world/2026/nov/02/supermarket-freezers-across-europe-hit-by-second-suspected-cyber-attack Comments URL: <a href="http...

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https://www.theguardian.com/world/2026/nov/02/supermarket-freezers-across-europe-hit-by-second-suspected-cyber-attack

Never Complain, Never Explain

👤 gregsadetsky 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://en.wikipedia.org/wiki/Never_complain,_never_explain Comments URL: https://news.ycombinator.com/item?id=50335659 Points: 4 <p...

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Private Jet plane crashes in Florida

👤 hmokiguess 📅 2026-11-02 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.cbc.ca/news/world/florida-plane-crash-9.7404996 Comments URL: https://news.ycombinator.com/item?id=50335914 Points: 6 <p...

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T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

Deidre Artis | 2026-09-11 | 1 min | 14 words | EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

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https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Analysis of the dynamic evolution and immunosuppressive functions of myeloid cells in gliomas based on single-cell transcriptomics

1 min | 23 words | BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Hao Yuan, Jingyi Yang, Zili Bai, Hongyan Wang, Jiaxuan Tong, Peng Chen, Shouping Gong

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600466X?dgcid=rss_sd_all

Common phenomenal and neural substrate geometry in visual motion perception

📖 1 min 📝 18 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Kallum Robinson, Ariel Zeleznikow-Johnston, Jiahao Wu, Yumiko Yoshimura, Naotsugu Tsuchiya

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https://www.sciencedirect.com/science/article/pii/S1053811926006282?dgcid=rss_sd_all

Neuronal and glial transcriptional states vary with proximity to tangle-bearing neurons in the human entorhinal cortex

👤 Boulger, S. L., Lambie, N., Wos, J., Papageorgopoulou, M., Mendez Pastor, S., Adair, E., Thomas, M., Caramello, A., Fancy, N. N., Jackson, J. S., Duff, E. P., Mallach, A., Matthews, P. M.

📅 2026-11-01 📖 1 min 📝 150 words BIORXIV NEUROSCIENCE

> **Summary:** More neurons die in Alzheimer's disease than form tangles, and tangle-bearing neurons can persist for decades, so tangles alone cannot explain degeneration. We combined 6,000-plex spatial transcriptomics with PHF1 immunofluorescence (167,176 cells; 9 donors) and imaging mass cytometry (170,195 cells...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.762234v1?rss=1

Microglial NUS1 deficiency drives dopaminergic neurodegeneration through intercellular iron dysregulation in Parkinson’s disease models

Haigang Ren 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02499-3 Using cellular and mouse models, this study reports that deficiency of the Parkinson’s disease risk gene NUS1 in microglia disrupts intercellu...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02499-3

Chronic progressive demyelination induces a microglial state with lysosomal dysfunction

Mikael Simons 2026-10-26 1 min 39 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 26 October 2026; doi:10.1038/s41593-026-02489-5 Acute myelin injury responses are well defined, but microglia adaptations to chronic damage remain unclear. Mouse models reveal a chronic demy...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02489-5

Calcium permeability–enhancing mutations in GluRIIC suppress sustained presynaptic homeostatic plasticity triggered by loss of GluRIIA

Olfat A. MalakLindsay GrayAmmar AlyRobin W. BallJessica Gustin-MurrayMyriam MoujahidineGuadalupe CarrascoAndrew RoskoEhud Y. IsacoffA. Pejmun Haghighiahttps://ror.org/050sv4x28Buck Institute for Research on Aging, Novato, CA 94945bhttps://ror.org/04b6nzv94Brigham and Women's Hospital, Harvard Medical School, Boston, MA 02115cInscopix Inc., Mountain View, CA 94043dhttps://ror.org/03taz7m60Leonard Davis School of Gerontology, University of Southern California, Los Angeles, CA 90089ehttps://ror.org/01pxwe438Department of Physiology, McGill University, Montreal, QC H3G1Y6, Canadafhttps://ror.org/01an7q238Helen Wills Neuroscience Institute, University of California, Berkeley, CA 94720ghttps://ror.org/01an7q238Department of Neuroscience, University of California, Berkeley, CA 94720hhttps://ror.org/01an7q238Department of Molecular and Cell Biology, University of California, Berkeley, CA 94720

📅 2026-10-27 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAccumulating experimental evidence indicates that neurons adjust their synaptic strength, in response to genetic or environmental/ pharmacological perturbations, to ensure stability in neural circu...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625954123?af=R

VAP organizes presynaptic Ca2+ signaling and synaptic vesicle mobilization through Kv2 membrane contact sites

👤 Cameron D. PatonIn Ha ChoMichael B. Hoppaahttps://ror.org/049s0rh22Division of Arts and Sciences, Department of Biological Sciences, Dartmouth College, Hanover, NH 03755

📅 2026-10-27 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThe endoplasmic reticulum (ER) supports neuronal function by regulating Ca2+, lipid, and protein homeostasis, yet how ER–plasma membrane contact sites contribute to synaptic transmission remains p...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2602106123?af=R

Intraventricular clindamycin enhances interstitial brain fluid drainage and the blood–brain barrier after a traumatic brain injury

👤 Landon S. AshbyMegan G. CroweDaveraj ChakravartySanjeev MedamGabriel FramptonAyesha B. DasDavid A. HormuthThomas E. YankeelovVincent Y. C. WangLinda J. Noble-HaeussleinBenjamin J. Umlaufahttps://ror.org/00hj54h04Department of Neurosurgery, The University of Texas at Austin, Austin, TX 78712bhttps://ror.org/00hj54h04Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712chttps://ror.org/00hj54h04Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX 78712dhttps://ror.org/00hj54h04Cancer Research Center, The University of Texas at Austin, Austin, TX 78712ehttps://ror.org/00hj54h04Department of Diagnostic Medicine, The University of Texas at Austin, Austin, TX 78712fDepartment of Imaging Physics, The University of Texas M.D. Anderson Cancer Center, Houston, TX 77030ghttps://ror.org/00hj54h04Department of Neurology, The University of Texas at Austin, Austin, TX 78712hhttps://ror.org/00hj54h04Department of Psychology, The University of Texas at Austin, Austin, TX 78712ihttps://ror.org/00hj54h04Mulva Clinic for the Neurosciences, The University of Texas at Austin, Austin, TX 78712

📅 2026-10-27 📖 1 min 📝 44 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDespite advances in the acute clinical management of severe traumatic brain injuries, treating injury-derived secondary pathogenesis remains a major challenge. We describe an alternative strategy ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2536755123?af=R

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

MTDANet: lightweight multi-scale temporal hierarchy and distance-regularized linear attention network for motor imagery brain–computer interfaces

Binbin Chen, Chen Deng, Huanmin Ge, Xinhua Su and Yanfei Shen 2026-10-26 1 min 239 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Motor-imagery (MI) electroencephalography (EEG) offers a non-invasive route to brain–computer interaction, yet reliable single-trial decoding remains difficult because task-related patterns are weak, nonstationary, and distributed over multiple temporal scales. This study investigates whe...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb330

AOR: adaptive orthogonal regression for depression feature selection with incomplete EEG channels

👤 Wenjia Dong, Zhijian Gong, Xueyuan Xu, Ke Liu, Fulin Wei and Li Zhuo

📅 2026-10-26 📖 1 min

📝 272 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Depression induces severe mental and physical impairments, highlighting the clinical need for rapid and reliable detection. Electroencephalography (EEG) directly reflects neural electrical activity and serves as a promising tool for depression investigation. However, the inherent non-stat...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aeb32f

Multimodal regulation of neurite extension by extracellular matrix composition and electrical stimulation on aligned collagen fiber scaffolds

👤 Madalynn J Thompson, Joanna Borowska-Fielding, Ivy A Murphy, Ying Zhang, Victor Rafuse and John P Frampton

📅 2026-10-26 📖 1 min 📝 279 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Peripheral nerve regeneration is regulated by structural, biochemical, and electrical cues, yet their combined effects on neurite outgrowth remain unclear. In this study, an aligned fibrous scaffold was developed to evaluate how extracellular matrix (ECM) composition and electrical stimul...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt

📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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